

LAM TVRMS — Zero to Production Project Guide

Oct 8, 2026 · @Syed Rahat

1. The project in one page

TVRMS (TV Request Management System) is one Power Platform solution that replaces the emails, PowerPoint files and spreadsheets Lam uses today for Test Vehicle (TV) requests. HCLTech builds it in 9 weeks at a fixed price of USD 58,950. Every request moves through 6 phases, from submission to closure.

Today vs. with TVRMS

Today (manual)	With TVRMS
Requests arrive by email, PowerPoint and Excel	One form in a Power Apps app, with mandatory fields and a mandatory technical document
No common request number	Every request gets a unique TVID, for example APTV_2026_014
Files are spread across mailboxes	One SharePoint folder per TVID with 5 standard sub-folders
Funding approval through long email chains	The Cost Center Owner approves in Outlook or Teams; the decision and comments are saved
Nobody knows the current status	8 clear statuses, automatic emails and reminders, and a Power BI dashboard
No audit trail	Every change is logged and kept for 7 years

Who uses it

Role	Who (sample person)	Main job in the app
Requester	Ravi Sharma, Lam engineer	Submits a request, answers questions, tracks progress
Additional Requester	Meena Iyer	Sees and updates the same request as the requester
APPG Owner	Priya Nair	Reviews the request, compares sourcing options, runs execution and closure
APPG Admin	Anand Rao	Assigns owners, holds or resumes requests, maintains master data
Cost Center Owner	Suresh Kumar (Level 1); Kavita Menon, Finance Head (Level 2)	Approves, rejects or holds the funding
RI Stakeholder	Rahul Verma, Lam contact for IMEC	Follows execution with the research institution
System Admin	HCLTech or Lam IT support	Monitors flows, fixes errors, deploys new versions

How your two documents fit together

The deck (ABC_REQUEST.pptx) is the **what**: the scope, architecture, timeline and price that Lam has seen. The build guide (LAM_TVRMS_Zero_to_Production_Build_Guide.docx) is the **how**: business rules, statuses, screens, flows, emails and tests. They disagree in 5 places; this guide follows the deck, keeps every business rule from the build guide, and lists each difference as a decision in Section 14.

Topic	Deck says	Build guide says	This guide follows	Decision
Where data is stored	Dataverse, plus a SharePoint document library	SharePoint lists only	Dataverse for data, SharePoint for documents	D1
Admin screens	Model-driven app for admins	Admin screen inside the canvas app	Model-driven app TVRMS Admin	D2

Topic	Deck says	Build guide says	This guide follows	Decision
Timeline	9 weeks with hypercare; go-live at the end of week 7	About 10 weeks to go-live, then 2 weeks of hypercare	The 9-week plan, with an MVP fallback	D3
Data migration and training	Out of scope	Chapter 13 (migration) and 15.1 (training)	Out of scope; offer as a change request	D4
Document workspace per TVID	Workspace per TVID	Folder per TVID with 5 sub-folders	One folder per TVID with 5 sub-folders	D5

How to read this guide

1. Sections 1 to 8 explain the solution: the story, architecture, roles, lifecycle, data, screens and flows.
2. Sections 9 to 13 are the build steps, in the order you do them: setup, data, apps, flows, security, testing, production.
3. Sections 14 to 17 are for the client: gaps to ask, assumptions, the plan and a short talk track.
4. Where the build guide already has the exact detail (email texts, user guides, HTTP calls), this guide points to that chapter instead of repeating it.

Words used in this guide

Word	Simple meaning
TV (Test Vehicle)	A test item Lam needs, such as test wafers, coupons, materials or modules
TVID	The unique request number, format APTV_YYYY_NNN (example APTV_2026_014)
APPG	The Lam team that reviews TV requests, finds sourcing options and runs execution
RI (Research Institution)	An institute, university or partner that can make or test the TV (example IMEC)

Word	Simple meaning
Feasibility	Comparing sourcing options by capability, cost, timeline and constraints
Dataverse	Microsoft's secure database for Power Platform; it stores all request data
Canvas app	The app end users work in; we design every screen
Model-driven app	An admin app generated from Dataverse tables, with ready-made forms and views
Solution	A package that holds the apps, flows, tables and settings; it moves from Dev to Test to Prod
Managed solution	A locked copy imported into Test and Prod; nobody edits it there
Environment variable	A setting with a different value in each environment (example: the SharePoint site URL)
Connection reference	A pointer to a connection (example: Outlook) that you choose at import time
DLP policy	Lam IT's rule on which connectors may be used together

2. Example use case: one request from start to end

Ravi Sharma needs 25 test wafers for an etch study. His request APTV_2026_014 goes through all 6 phases between 07-Oct-2026 and 12-Dec-2026. Use this story in demos and whenever you explain the system to Lam; every section of this guide uses the same sample data.

The sample request

Field	Sample value
TVID	APTV_2026_014
Request Title	25 test wafers – 300mm etch study

Field	Sample value
Request Type / Frequency	New / One-time
TV Type / Quantity	Wafer / 25
Description	300mm wafers with 20nm oxide layer for etch tests
Business Justification	Needed for Q1 product qualification
Need By Date	15-Dec-2026
Cost Center	CC-4410 (Etch R&D)
Estimated Budget	USD 18,000
Requester / Additional Requester	Ravi Sharma / Meena Iyer
Document	Wafer_Spec_v1.pdf (Technical Documentation Package)

The story, step by step

Date (sample)	Phase	Who does what	What the system does	Status after	Email
07-Oct	1. Request Initiation	Ravi fills the form, attaches Wafer_Spec_v1.pdf, adds Meena and clicks Submit	Checks mandatory fields; saves the row; gives TVID APTV_2026_014; creates the folder with 5 sub-folders; saves the PDF in 01_Request; shares the request with Meena	Submitted	N01 to Ravi (cc Meena); N02 to the APPG team
08-Oct	2. APPG Review	Priya sees the budget breakdown is missing and clicks Mark Missing Information	Logs the change and starts the 30-day clock	Missing Information	N04 to Ravi
09-Oct	2. APPG Review	Ravi uploads Budget_Breakdown.xlsx and clicks Edit and Resubmit	Stops the clock and logs the change	Submitted	N09 to the APPG team

Date (sample)	Phase	Who does what	What the system does	Status after	Email
09-Oct	2. APPG Review	Anand assigns Priya as the owner	Stamps the feasibility start date	Sourcing Feasibility In Progress	N03 to Ravi (cc Priya)
10-14 Oct	3. Internal Alignment, round 1	Priya adds 2 options: IMEC (USD 18,000, 8 weeks, Recommended) and IIT Bombay (USD 11,000, 12 weeks, Not Recommended)	Saves both options under the request	Sourcing Feasibility In Progress	–
15-Oct	3. Internal Alignment, round 2	Priya selects IMEC, generates the evaluation report and clicks Send for funding approval	Creates the evaluation report PDF in 02_Feasibility; reads CC-4410 in the approval matrix and finds Suresh	Pending Funding Approval	N10 to Priya (cc Ravi); N11 approval card to Suresh
16-Oct	4. Funding Approval	Suresh clicks Approve in Outlook with the comment "OK within budget"	18,000 is not above the 50,000 limit, so no Level 2; saves the approval history	Pending Funding Approval (Approval Status = Approved)	N14 to Ravi (cc Priya)
20-Oct	5. Execution	Priya clicks Start execution and adds 4 milestones	Shares the request with Rahul Verma (RI Stakeholder for IMEC)	Execution In Progress	N17 to Ravi (cc Priya, Rahul)
18-Nov	5. Execution	The milestone Interim results review is due on 20-Nov	Sends the reminder 2 days before the due date	Execution In Progress	N18 to Priya
12-Dec	6. Closure	Priya uploads Final_Report.pdf, enters the actual cost USD 17,200, the outcome and lessons learned, and clicks Close request	Stamps the closed date; keeps the record and files for 7 years	Request Complete	N19 to Ravi (cc Meena, Priya, Suresh)

Note: on 09-Oct the request had no owner yet, so N09 went to the APPG team. Once an owner is assigned, N09 goes to the owner.

What happens in other cases

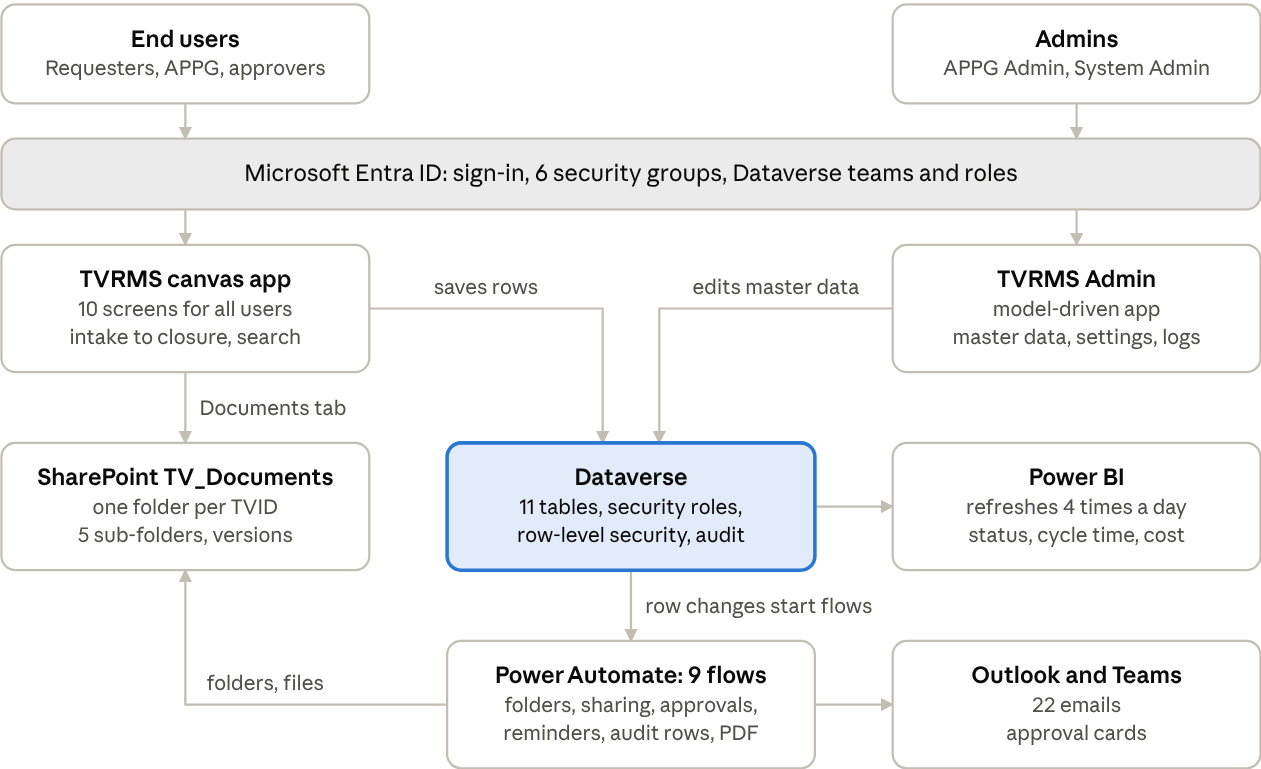
- **Ravi never answers the question:** reminders go out on day 7, 14 and 25 (N05, N06, N07). On day 30 the request is cancelled automatically (N08) and stays in the system for history.
- **The amount is USD 60,000:** Suresh approves first (Level 1), then Kavita approves (Level 2), because 60,000 is above the 50,000 limit of CC-4410.
- **Suresh chooses Hold:** the status becomes Sourcing On Hold (N16) until Anand resumes the approval or cancels the request.
- **Suresh rejects:** the status becomes Request Cancelled (N15). The record is kept; nothing is ever deleted.
- **Priya finds the request is not a TV request:** she clicks Reject at review with a reason. The status becomes Request Cancelled with the category Rejected by APPG (N20).
- **Ravi needs the same wafers next quarter:** he opens Search History, picks APTV_2026_014 and clicks Copy as Repeat Request. The form is pre-filled, Request Type = Repeat, and the new request (example APTV_2027_005) points to APTV_2026_014 as its parent.

3. Solution architecture

TVRMS uses 5 layers inside Lam's own Microsoft 365 tenant. People sign in with Entra ID and work in 2 apps; data lives in Dataverse and files in SharePoint; 9 flows do the background work; Power BI shows the reports.

Two apps share one Dataverse; 9 flows do the background work

Everything stays inside the Lam Microsoft 365 tenant



solution architecture · 5 layers, 9 components

Arrows show who talks to whom: people sign in through Entra ID, both apps save to Dataverse, and every row change starts a flow that writes to SharePoint and sends emails and approval cards. The canvas app also calls flows F08 and F09 directly, to make the PDF report and to upload files.

What each part does

Layer	Component	What it does in TVRMS	Example
Identity	Microsoft Entra ID	Sign-in; 6 security groups decide each person's role	Priya is in TVRMS-APPG-Owners
User interface	Canvas app TVRMS	10 screens for requesters, APPG and approvers; validation; report button	Ravi submits his request

Layer	Component	What it does in TVRMS	Example
User interface	Model-driven app TVRMS Admin	Master data, settings, logs and audit history for admins	Anand adds cost center CC-7020
Data	Dataverse (11 tables)	Request data, security roles, row-level security, audit	The row APTV_2026_014
Documents	SharePoint library TV_Documents	One folder per TVID, 5 sub-folders, versioning, 7-year retention	/TV_Documents/APTV_2026_014/01_Request
Automation	Power Automate (9 flows)	Folders, sharing, approvals, reminders, emails, audit rows, PDF	F03 sends the approval card
Notifications	Outlook and Approvals (also shown in Teams)	22 emails and the approval cards	N11 reaches Suresh
Reports	Power BI	Status, cycle time, aging, cost, exceptions	Average cycle time by month

How one request moves through the layers

1. Ravi opens the canvas app. Entra ID signs him in, and Dataverse applies his security role.
2. He submits the form. The app saves one row in the Dataverse table TV Request, and Dataverse gives the TVID at once. The app sends his files to flow F09, which saves them in SharePoint.
3. Flow F01 sees the new row. It creates the TVID folder, sets folder permissions, shares the row with Meena, writes the first audit row and sends N01 and N02.
4. Every status change by Priya starts flow F02. It writes an audit row and sends the right email.
5. When Priya sends the request for funding, flow F03 finds the approver in the Cost Center table and sends the approval card to Suresh in Outlook and Teams.

6. Daily flows (F04, F05, F06) send reminders and auto-cancel old requests. Power BI refreshes from Dataverse 4 times a day.

Licences needed (confirm with Lam IT)

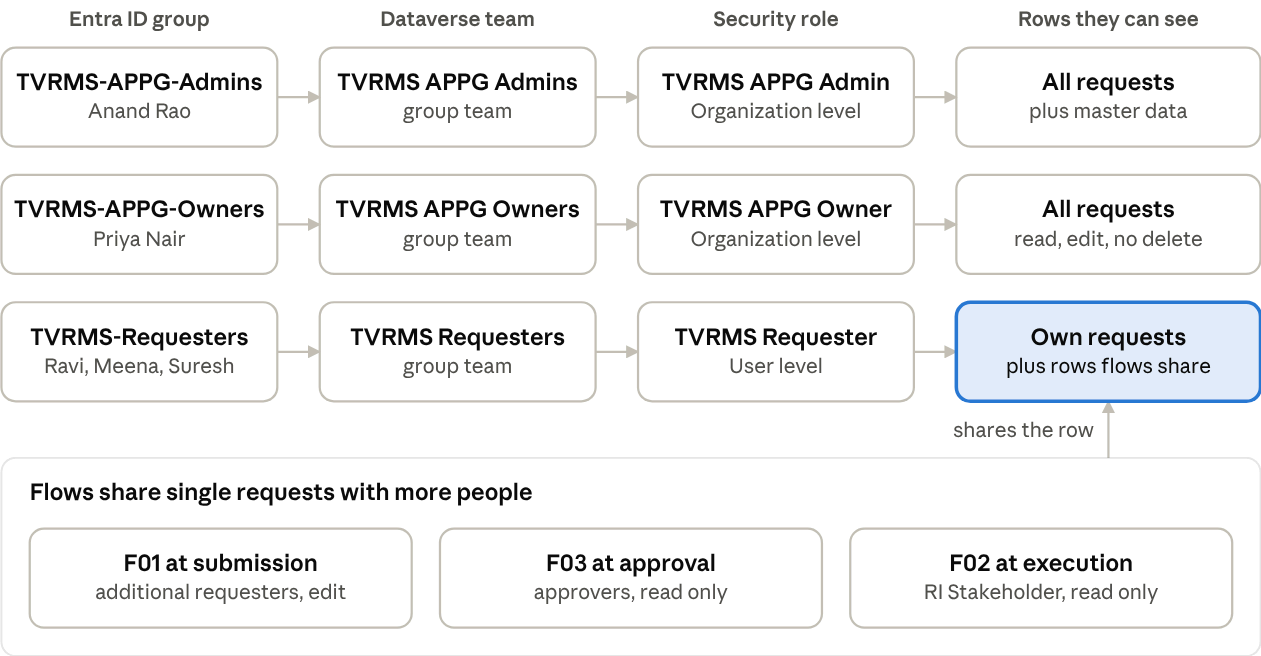
Who	Licence	Why
Every app user (requesters, APPG, admins)	Power Apps Premium (per user) or a per-app plan	Both apps use Dataverse, which is a premium data source
Approvers who only act in Outlook or Teams	Microsoft 365 only	They never open the app; if they also use My Approvals, they need an app licence
Service account svc-tvrms	Microsoft 365 with a mailbox, plus Power Automate Premium or a Process licence	The background flows use the premium Dataverse connector
Power BI viewers (APPG, management)	Power BI Pro, or Premium/Fabric capacity	Deck assumption: Lam provides these
You (developer)	Power Apps Premium, or the free Developer Plan for building	To build and test in Dev

Licensing rules change often, so ask Lam's licensing team to confirm this table before week 3 (decision D1).

4. Roles and permissions

Seven roles decide what each person can see and do. Dataverse security roles enforce this on the server, and the app only hides buttons a person should not use. So security never depends on the app alone: a requester who opens Dataverse directly still sees only his own requests.

Access comes from group, team and role; flows add access per request



security model · group, team, role and sharing

Read each row left to right: IT puts a person in an Entra group, the linked Dataverse team gives the security role, and the role decides which rows the person sees. Requesters see only their own requests until a flow shares one with them. System Admins and the service account are left out of the picture to keep it simple.

Who can do what

Action	Requester	Additional Requester	APPG Owner	APPG Admin	RI Stakeholder	Cost Center Owner	System Admin
Submit a new request	Yes	Yes	Yes	Yes	Yes	Yes	–
See own and linked requests	Yes	Yes	Yes	Yes	After execution starts	While approving	Yes
See all requests	No	No	Yes	Yes	No	No	Yes
Edit and resubmit (Missing Information)	Yes	Yes	No	Yes	No	No	No

Action	Requester	Additional Requester	APPG Owner	APPG Admin	RI Stakeholder	Cost Center Owner	System Admin
Cancel a request	While Submitted or Missing Information	No	No	Any open status	No	No	No
Assign APPG owner	No	No	Yes	Yes	No	No	No
Mark Missing Information or reject at review	No	No	Yes	Yes	No	No	No
Add options, make the report, send for approval	No	No	Yes	Yes	No	No	No
Approve, reject or hold funding	No	No	No	No	No	Yes	No
Resume approval after Hold	No	No	No	Yes	No	No	No
Execution, milestones, issues, closure	No	No	Yes	Yes	No	No	No
See audit and approval history	No	No	Yes	Yes	No	No	Yes
Edit master data and settings	No	No	No	Yes	No	No	Yes
Monitor flows and deploy	No	No	No	No	No	No	Yes

How each role is built

For each Entra security group you create one Dataverse team of type Microsoft Entra ID Security Group, and you give that team a security role. When IT adds a person to the group, the person gets the role automatically.

Role	Entra security group	Security role (given to the team)	TV Request rows the person can see
Requester, Additional Requester	TVRMS-Requesters	TVRMS Requester	Own rows, plus rows shared with them
Cost Center Owner	TVRMS-Requesters (and TVRMS-CostCenter-Owners for reviews)	TVRMS Requester	Own rows, plus rows F03 shares for approval
RI Stakeholder	TVRMS-Requesters (and TVRMS-RI-Stakeholders for reviews)	TVRMS Requester	Own rows, plus rows F02 shares at execution start
APPG Owner	TVRMS-APPG-Owners	TVRMS APPG Owner	All rows (read and edit, no delete)
APPG Admin	TVRMS-APPG-Admins	TVRMS APPG Admin	All rows, plus master data and settings
System Admin	TVRMS-System-Admins	System Administrator	Everything in the environment
Service account svc-tvrms	– (added as a user)	TVRMS Service	Everything in TVRMS; it runs the flows and shares rows

Security role privileges

"User" level means rows the person owns, rows shared with them and rows owned by their teams. "Organization" level means all rows. Create each role by copying the standard Basic User role, so the app's basic system tables keep working.

Table	TVRMS Requester	TVRMS APPG Owner	TVRMS APPG Admin	TVRMS Service
TV Request	Create, Read, Write, Append, Append To: User	Create, Read, Write, Append, Append To: Organization	Same as APPG Owner, plus Share	All except Delete: Organization, incl. Share
Feasibility Option, Execution Milestone, TV Issue	None	Create, Read, Write, Append, Append To: Organization	Same as APPG Owner	All except Delete: Organization
Status History, Approval History, Notification Log	None	Read: Organization	Read: Organization	Create, Read: Organization
Cost Center, Research Institution, APPG Stakeholder, TVRMS Setting	Read: Organization	Read: Organization	Create, Read, Write: Organization	Read: Organization
Delete on any TVRMS table	None	None	None	None

Nobody can delete. Old master data is switched off with Is Active = No, and records stay for the 7-year retention.

Extra access given by flows

When	Flow	Who gets access	Access
Request submitted	F01	The requester (owner of the row)	Read on the TVID folder
Request submitted	F01	Additional requesters	Read and Write on the row; Read on the TVID folder

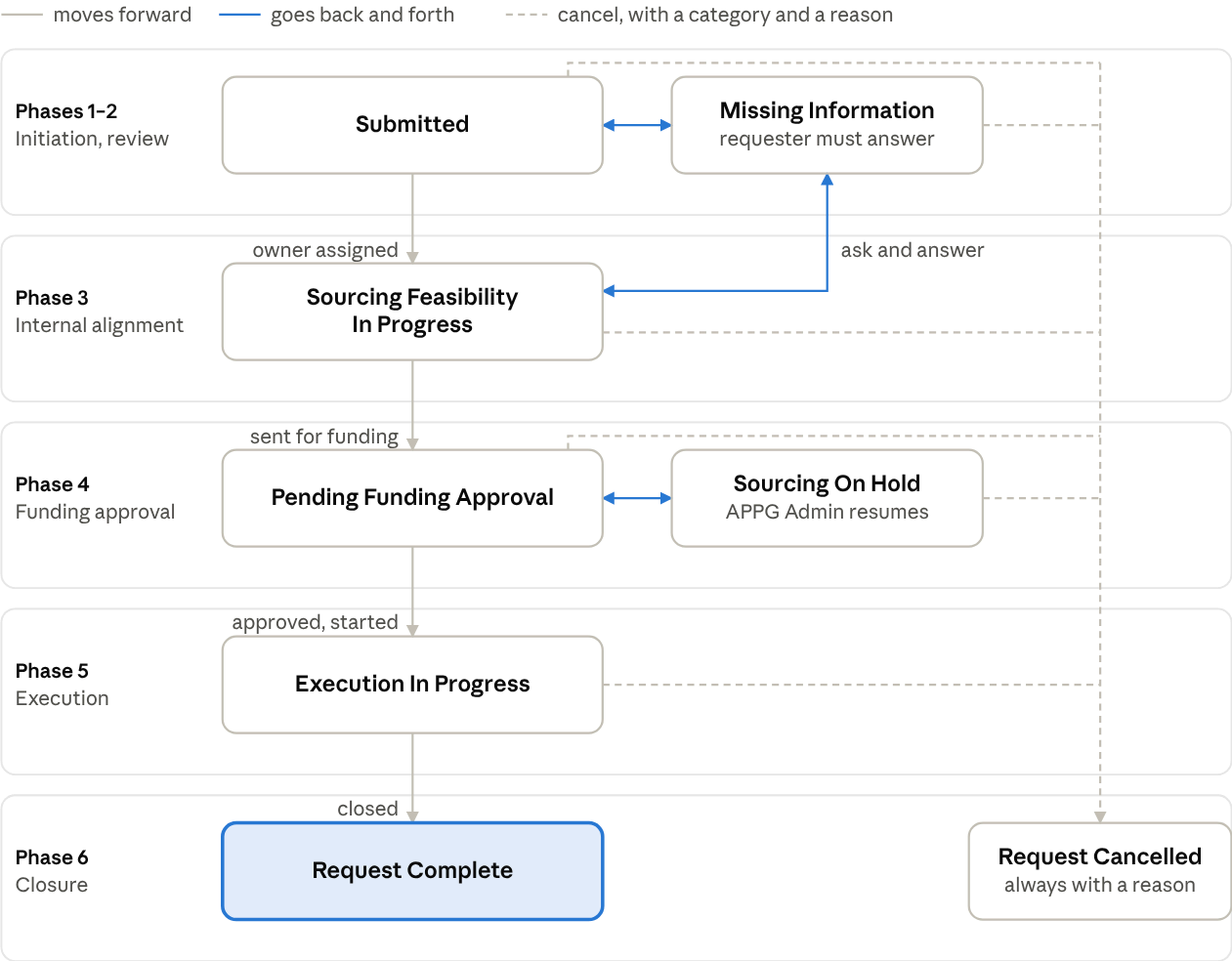
When	Flow	Who gets access	Access
Request submitted	F01	SharePoint groups TVRMS APPG Owners and Admins	Contribute No Delete on the TVID folder
Sent for funding approval	F03	Level 1 (and Level 2) approver	Read on the row and the folder
Execution started	F02	RI Stakeholder of the selected institution	Read on the row and the folder

Example: Ravi and Meena can open APTV_2026_014; another engineer cannot see it in the app, in Dataverse or in SharePoint. Suresh sees it only after it is sent to him for approval. Ravi can still add files later, because flow F09 saves them using the service account.

5. Request lifecycle: 6 phases, 8 statuses

Every request has exactly one of 8 statuses at any time. The 6 phases in the deck are groups of these statuses: the business talks in phases, and the system tracks statuses.

A request moves down 6 phases; any open status can be cancelled



request lifecycle · 8 statuses in 6 phases

Read it top to bottom. Grey arrows are the normal path, the coloured two-way arrows are the go-back moves (missing information, hold and resume), and the dashed lines show that every open status can end in Request Cancelled.

Phases and statuses

Phase (deck)	Status in the system	Main screen	Who acts
1. Request Initiation	Submitted	New Request	Requester
2. APPG Review	Submitted, Missing Information	APPG Queue, Request Detail	APPG Owner or Admin, Requester

Phase (deck)	Status in the system	Main screen	Who acts
3. Internal Alignment (round 1 and round 2)	Sourcing Feasibility In Progress	Feasibility	APPG Owner
4. Funding Approval	Pending Funding Approval, Sourcing On Hold	Approval card in Outlook or Teams, My Approvals	Cost Center Owner, APPG Admin
5. Execution	Execution In Progress	Execution	APPG Owner
6. Closure	Request Complete (or Request Cancelled from any open status)	Closure	APPG Owner

Allowed moves

From	To	Who	How
(new)	Submitted	Requester	New Request → Submit
Submitted	Sourcing Feasibility In Progress	APPG Owner or Admin	Request Detail → Assign owner
Submitted or Sourcing Feasibility In Progress	Missing Information	APPG Owner or Admin	Request Detail → Mark Missing Information
Missing Information	Submitted (no owner yet) or Sourcing Feasibility In Progress (owner assigned)	Requester	Request Detail → Edit and Resubmit
Missing Information	Request Cancelled	System	Flow F04 after 30 days with no update
Submitted	Request Cancelled	APPG Owner or Admin	Request Detail → Reject at review (reason required)

From	To	Who	How
Sourcing Feasibility In Progress	Pending Funding Approval	APPG Owner	Feasibility → Send for funding approval
Pending Funding Approval	Sourcing On Hold	Cost Center Owner	Approval card → Hold (flow F03)
Pending Funding Approval	Request Cancelled	Cost Center Owner	Approval card → Reject (flow F03)
Sourcing On Hold	Pending Funding Approval	APPG Admin	Request Detail → Resume approval
Pending Funding Approval, with Approval Status = Approved	Execution In Progress	APPG Owner	Execution → Start execution
Execution In Progress	Request Complete	APPG Owner	Closure → Close request
Any open status	Request Cancelled	APPG Admin; the requester only while Submitted or Missing Information	Request Detail → Cancel (reason required)

Three rules to remember

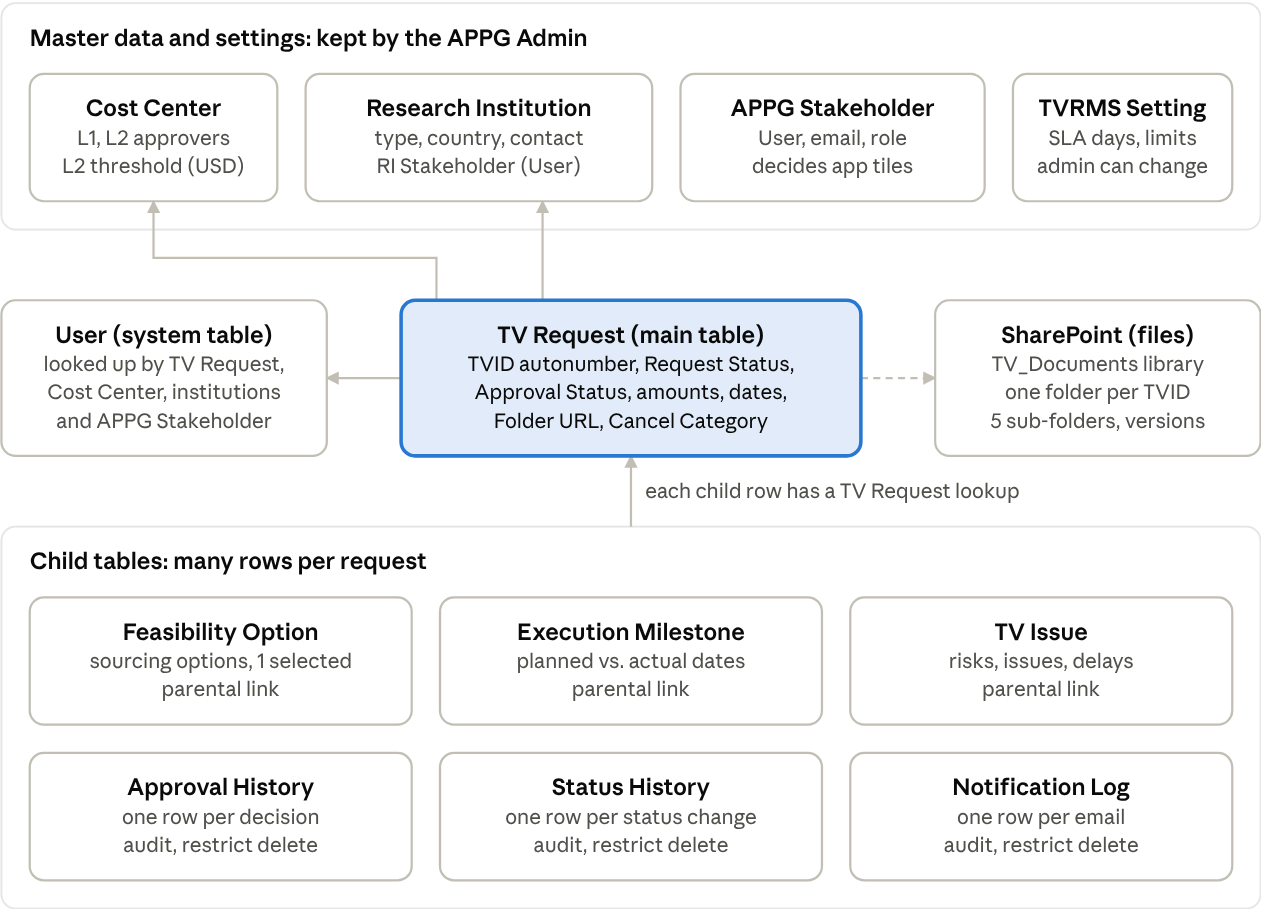
1. **Approved is not a status.** After approval the status stays Pending Funding Approval and the column Approval Status becomes Approved. This unlocks Start execution and keeps exactly the 8 statuses.
2. **Execution guard.** If anyone sets Execution In Progress before approval, even directly in Dataverse, flow F02 puts the old status back and alerts the APPG team.
3. **Every cancellation has a category and a reason.** Cancel Category is one of: Cancelled by requester, Rejected by APPG, Funding rejected, Auto-cancelled, Cancelled by admin. This feeds the exception reports in Power BI.

6. Data model

All request data lives in 11 Dataverse tables, and all files live in one SharePoint library. TV Request is the centre: 6 child tables hang under it, and 4 master tables feed it.

TV Request is the hub: 6 child tables below, 4 master tables above

Arrows run from the table holding the lookup to the table it looks up (many to one); dashed = linked by name



data model · 11 tables and 1 SharePoint library

Read each arrow as looks up: TV Request looks up its cost center, its institution and its people, and every child row looks up its TV Request. Files are not stored in Dataverse; each request's Folder URL points to its SharePoint folder.

The tables at a glance

#	Table (schema name)	Kind	Ownership	One row means	Who writes
1	TV Request (lam_tvrequest)	Main	User or team	One TV request	App and flows
2	Feasibility Option (lam_feasibilityoption)	Child	User or team	One sourcing option for a request	APPG (app)
3	Execution Milestone (lam_executionmilestone)	Child	User or team	One planned step during execution	APPG (app)

#	Table (schema name)	Kind	Ownership	One row means	Who writes
4	TV Issue (lam_tvissue)	Child	User or team	One risk, issue or exception during execution	APPG (app)
5	Approval History (lam_approvalhistory)	Child, audit	Organization	One approval decision	Flow F03 only
6	Status History (lam_statushistory)	Child, audit	Organization	One status change	Flows only
7	Notification Log (lam_notificationlog)	Child, audit	Organization	One email sent	Flows only
8	Cost Center (lam_costcenter)	Master	Organization	One cost center and its approvers (the approval matrix)	APPG Admin
9	Research Institution (lam_researchinstitution)	Master	Organization	One institute, university or partner	APPG Admin
10	APPG Stakeholder (lam_appgstakeholder)	Master	Organization	One APPG person and their role	APPG Admin
11	TVRMS Setting (lam_tvrmssetting)	Settings	Organization	One business setting, such as an SLA day count	APPG Admin
L1	TV_Documents (SharePoint library)	Documents	-	One folder per TVID	Flows (and APPG)
L2	TV_Templates (SharePoint library)	Documents	-	Logo, user guides, report template	APPG Admin

Compared with the build guide, its 10 SharePoint lists become these tables. TVID_Counter is not needed, because Dataverse makes the number. TV Issue and TVRMS Setting are new: they cover the deck's exception management and its rule that configuration changes must be auditable.

Important: you cannot change a table's ownership after you create it. Requests and their working child tables are User or team owned, so row-level security and sharing work. Masters and logs are Organization owned, because their access is all or nothing.

How the tables are related

Parent (one)	Child (many)	Lookup column on the child	Behaviour	Why
TV Request	Feasibility Option, Execution Milestone, TV Issue	TV Request	Parental	They belong to the request and follow it
TV Request	Approval History, Status History, Notification Log	TV Request	Referential, restrict delete	Audit rows can never lose their request
TV Request	TV Request (a repeat)	Parent Request	Referential	A repeat request points to the earlier one
Cost Center	TV Request	Cost Center	Referential, restrict delete	A cost center in use cannot be removed
Research Institution	Feasibility Option; TV Request	Research Institution; Selected Institution	Referential	Options and the final choice point to the master list
Feasibility Option	TV Request	Selected Option	Referential	The request points to the chosen option
User (system table)	TV Request; Cost Center; Research Institution; APPG Stakeholder	Requester, Assigned APPG Owner, Current Approver; Level 1 and Level 2 Approver; RI Stakeholder; User	Referential	People are real Lam users, so sharing and emails just work
TV Request	TV_Documents folder	TVID (the folder name) and Folder URL	Linked by name, not a lookup	Files stay in SharePoint, as the deck requires

Example: request APTV_2026_014 ends with 2 feasibility options, 4 milestones, 1 issue, 1 approval history row, 7 status history rows and about 10 notification log rows.

Table 1 – TV Request (main table)

Column (display name)	Type	Required	Filled by	Sample value
Request Title (primary column)	Text (200)	Yes	Requester	25 test wafers – 300mm etch study
TVID	Autonumber, format APTV_{DATETIMEUTC:yyyy}_{SEQNUM:3}	–	Dataverse	APTV_2026_014
Request Type	Choice: New, Repeat	Yes	Requester	New
Parent Request	Lookup → TV Request	If Repeat	Requester	APTV_2025_087
Frequency	Choice: One-time, Recurring	Yes	Requester	One-time
Recurrence Details	Multiline text	If Recurring	Requester	Every quarter, 25 wafers
TV Type	Choice: Wafer, Coupon, Material, Module, Other	Yes	Requester	Wafer
Description	Multiline text	Yes	Requester	300mm wafers with 20nm oxide layer for etch tests
Business Justification	Multiline text	Yes	Requester	Needed for Q1 product qualification
Quantity	Whole number	Yes	Requester	25
Need By Date	Date only	Yes	Requester	15-Dec-2026
Cost Center	Lookup → Cost Center	Yes	Requester	CC-4410
Estimated Budget (USD)	Decimal, 2 places	Yes	Requester	18,000.00
Requester	Lookup → User	Yes	App (current user)	Ravi Sharma
Additional Requesters	Multiline text (emails separated by ;)	No	Requester	meena.iyer@lamresearch.com
Request Status	Choice (global TVRMS Request Status), default Submitted	Yes	App and flows	Submitted
Previous Status	Choice (global TVRMS Request Status), hidden helper	–	Flows F01, F02	Submitted
Assigned APPG Owner	Lookup → User	–	APPG	Priya Nair
Missing Info Comments	Multiline text	When asking	APPG	Please add the budget breakdown
Missing Info Since	Date and time	–	App	08-Oct-2026 09:00

Column (display name)	Type	Required	Filled by	Sample value
Feasibility Started On	Date and time	–	F02	09-Oct-2026 11:00
Selected Option	Lookup → Feasibility Option	–	APPG	Option A – IMEC
Selected Institution	Lookup → Research Institution	–	APPG	IMEC
Funding Amount (USD)	Decimal, 2 places	Before approval	APPG	18,000.00
Report Generated On	Date and time	–	F08	15-Oct-2026 09:40
Approval Status	Choice (global TVRMS Approval Status), default Not Started	–	App and F03	Approved
Current Approver	Lookup → User	–	F03	Suresh Kumar
Approval Requested On	Date and time	–	F03	15-Oct-2026 10:00
Approved On	Date and time	–	F03	16-Oct-2026 14:20
Execution Start Date	Date only	–	F02	20-Oct-2026
Expected Completion Date	Date only	At execution start	APPG	10-Dec-2026
Execution Progress (%)	Whole number, 0 to 100	–	APPG	60
Actual Cost (USD)	Decimal, 2 places	At closure	APPG	17,200.00
Outcome	Multiline text	At closure	APPG	All 25 wafers delivered and met spec
Lessons Learned	Multiline text	No	APPG	Book the lab slot 4 weeks early
Reuse Recommendation	Choice: Reuse as is, Reuse with changes, Do not reuse	At closure	APPG	Reuse with changes
Closed On	Date and time	–	F02	12-Dec-2026 16:00
Cancel Category	Choice: Cancelled by requester, Rejected by APPG, Funding rejected, Auto-cancelled, Cancelled by admin	When cancelling	App or flows	–
Cancel Reason	Multiline text	When cancelling	App or flows	–
Folder URL	URL	–	F01	https://lam.sharepoint.com/sites/TVRMS/TV_Documents/APTV_2026_014
Is Doc Uploaded	Yes/No, default No	–	F09	Yes

You also get system columns for free: Created On (used as the submitted date), Created By, Modified On, Modified By and Owner.

Table 2 – Feasibility Option

Column	Type	Option A	Option B
Option Name (primary column)	Text	Option A – IMEC	Option B – IIT Bombay
TV Request	Lookup → TV Request (parental)	APTV_2026_014	APTV_2026_014
Research Institution	Lookup → Research Institution	IMEC	IIT Bombay
Technical Capability	Choice: High, Medium, Low	High	Medium
Estimated Cost (USD)	Decimal, 2 places	18,000	11,000
Estimated Timeline (weeks)	Whole number	8	12
Constraints	Multiline text	Needs NDA before start	–
Quote Reference	Text	IMEC-Q-2026-118	IITB-Q-077
Recommendation	Choice: Recommended, Backup, Not Recommended	Recommended	Not Recommended
Comments	Multiline text	Best quality, slightly higher cost	Lower cost but longer timeline
Is Selected	Yes/No	Yes	No
Assessed By	Lookup → User	Priya Nair	Priya Nair

Table 3 – Execution Milestone

Columns: Milestone Name (primary), TV Request (lookup, parental), Owner (lookup to User), Planned Date and Actual Date (date only), Milestone Status (choice: Not Started, In Progress, Done, Delayed), Comments.

Milestone Name	Planned Date	Actual Date	Milestone Status	Owner
Purchase order raised	22-Oct-2026	22-Oct-2026	Done	Priya Nair
Wafer processing started	27-Oct-2026	29-Oct-2026	Done	Priya Nair
Interim results review	20-Nov-2026	–	In Progress	Priya Nair
Final wafers shipped	05-Dec-2026	–	Not Started	Priya Nair

Table 4 – TV Issue (new)

Columns: Issue Title (primary), TV Request (lookup, parental), Issue Type (choice: Risk, Issue, Delay, Cost overrun, Quality), Severity (choice: High, Medium, Low), Owner (lookup to User), Raised On and Target Date (date only), Issue Status (choice: Open, In Progress, Resolved), Resolution.

Issue Title	Issue Type	Severity	Raised On	Target Date	Issue Status	Resolution
Lab etch tool down for 1 week	Delay	Medium	03-Nov-2026	10-Nov-2026	Resolved	IMEC moved our slot to the next week

Table 5 – Approval History

Columns: Name (primary), TV Request (lookup), Approval Level (whole number), Approver (lookup to User), Decision (choice: Approved, Rejected, On Hold), Comments, Decision Date (date and time), Amount (USD).

Name	Approval Level	Approver	Decision	Comments	Decision Date	Amount (USD)
APTV_2026_014 – Level 1	1	Suresh Kumar	Approved	OK within budget	16-Oct-2026 14:20	18,000

Table 6 – Status History

Columns: Name (the TVID), TV Request (lookup), Old Status and New Status (global TVRMS Request Status), Changed By (lookup to User), Changed On (date and time), Comments.

Name	Old Status	New Status	Changed By	Changed On	Comments
APTV_2026_014	-	Submitted	Ravi Sharma	07-Oct-2026 10:30	Request submitted
APTV_2026_014	Submitted	Missing Information	Priya Nair	08-Oct-2026 09:01	Please add the budget breakdown
APTV_2026_014	Missing Information	Submitted	Ravi Sharma	09-Oct-2026 08:45	Budget breakdown added

Table 7 – Notification Log

Columns: Subject (primary), TV Request (lookup), Notification Code (text), Sent To (text), Sent On (date and time), Delivery Status (choice: Sent, Failed).

Subject	Notification Code	Sent To	Sent On	Delivery Status
TV request APTV_2026_014 submitted – 25 test wafers – 300mm etch study	N01	ravi.sharma@lamresearch.com; meena.iyer@lamresearch.com	07-Oct-2026 10:31	Sent

Table 8 – Cost Center (the approval matrix)

Cost Center Code	Cost Center Name	Level 1 Approver	Level 2 Approver	Level 2 Threshold (USD)	Is Active
CC-4410	Etch R&D	Suresh Kumar	Kavita Menon	50,000	Yes
CC-5120	Deposition R&D	Lakshmi Prasad	Kavita Menon	50,000	Yes
CC-6030	Metrology	Daniel Lee	Kavita Menon	75,000	Yes

Level 2 is needed only when the Funding Amount is above the Level 2 Threshold.

APTV_2026_014 asks USD 18,000 on CC-4410, so only Suresh approves; USD 60,000 would go to Suresh first and then to Kavita.

Table 9 – Research Institution

Institution Name	Institution Type	Country	Contact (external, information only)	RI Stakeholder (Lam user)	Is Active
IMEC	Research Institution	Belgium	Dr. Jan Peeters, jan.peeters@imec.be	Rahul Verma	Yes
IIT Bombay	University	India	Prof. A. Kulkarni, a.kulkarni@iitb.ac.in	Neha Joshi	Yes
CEA-Leti	Research Institution	France	Dr. Marie Dubois, marie.dubois@cea.fr	Rahul Verma	Yes

Contact Name and Contact Email are two text columns. Institution Type is a choice: Research Institution, University, Partner.

Table 10 – APPG Stakeholder

Full Name	User	Email (lowercase)	Role (global TVRMS Role)	Is Active
Priya Nair	Priya Nair	priya.nair@lamresearch.com	APPG Owner	Yes
Arjun Mehta	Arjun Mehta	arjun.mehta@lamresearch.com	APPG Owner	Yes
Anand Rao	Anand Rao	anand.rao@lamresearch.com	APPG Admin	Yes
Rahul Verma	Rahul Verma	rahul.verma@lamresearch.com	RI Stakeholder	Yes

The app reads this table at start-up to decide which tiles to show. Keep it in step with the Entra groups: a new APPG Owner goes into the group and into this table.

Table 11 – TVRMS Setting (new)

Setting Name	Number Value	Text Value	Used by
SLA_ReviewDays	5	–	F05: unassigned requests (N21)

Setting Name	Number Value	Text Value	Used by
SLA_ApprovalReminderDays	3	–	F05: approval reminder (N12)
SLA_ApprovalEscalationDays	6	–	F05: escalation (N13)
MissingInfoCancelDays	30	–	F04: auto-cancel (N08); final warning 5 days before
MilestoneReminderDays	2	–	F06: due-soon reminder (N18)
MaxUploadMB	10	–	App: largest file per upload
AdminAppURL	–	Link of TVRMS Admin	App: Admin tile

Global choices (shared by tables and formulas)

Choice	Values, in this order
TVRMS Request Status	Submitted, Missing Information, Sourcing Feasibility In Progress, Pending Funding Approval, Sourcing On Hold, Execution In Progress, Request Complete, Request Cancelled
TVRMS Approval Status	Not Started, Pending, Approved, Rejected, On Hold
TVRMS Role	APPG Owner, APPG Admin, RI Stakeholder, System Admin

Do not use the system columns Status and Status Reason for the request status. In Dataverse they mean Active or Inactive; our own column is Request Status.

SharePoint library TV_Documents

TV_Documents

└─ APTV_2026_014	
└─ 01_Request	Technical Documentation Package, requester updates
└─ 02_Feasibility	Evaluation report (PDF), quotes
└─ 03_Approval	Approval evidence
└─ 04_Execution	Progress reports
└─ 05_Closure	Final deliverables, final report, lessons learned

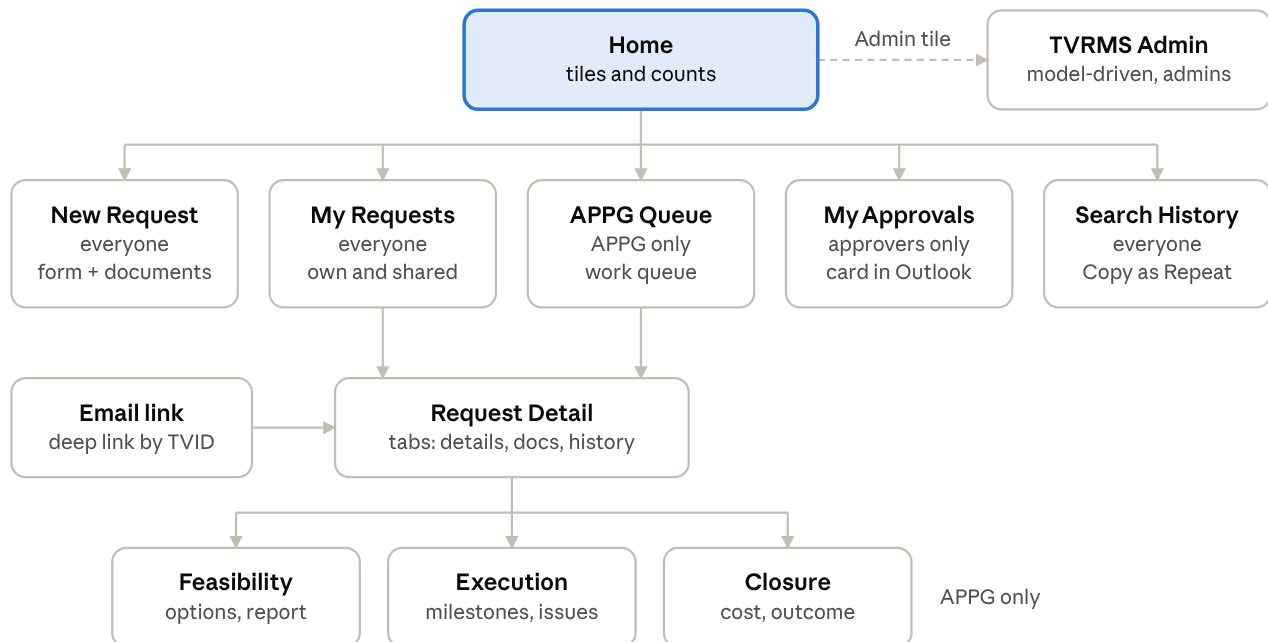
Library columns: TVID (single line of text, indexed), Document Category (choice: 01_Request to 05_Closure) and Uploaded By Email (single line of text). Versioning keeps 100 major versions. The evaluation report always keeps the same file name, so every older version stays available (v1, v2 and so on).

7. App screens

One canvas app with 10 screens serves every user, and the model-driven app TVRMS Admin serves the admins. The person's role decides which tiles and buttons appear, and the request status decides which actions are allowed.

Home opens every screen; the role decides which tiles appear

Canvas app TVRMS: 10 screens. Admins also get the model-driven app TVRMS Admin.



screen map · 10 canvas screens and the admin app

Everyone sees New Request, My Requests and Search History; APPG and approvers get extra tiles. Feasibility, Execution and Closure open from Request Detail, and every email link jumps straight to Request Detail.

Canvas app TVRMS - 10 screens

#	Screen	Who sees it	What it shows	Main buttons
1	scrHome	Everyone	Tiles and small counts: Need my action, Waiting for my approval, New in queue	New Request, My Requests, APPG Queue, My Approvals, Search History, Admin
2	scrNewRequest	Everyone	The request form and the document upload	Submit, Cancel
3	scrMyRequests	Everyone	Requests I created or that were shared with me; search and status filter	Open
4	scrRequestDetail	Everyone (buttons by role)	Tabs: Details, Documents, Approval, History	Edit and Resubmit, Cancel, Assign owner, Mark Missing Information, Reject at review, Resume approval, Upload document, Open folder
5	scrAPPGQueue	APPG	Work queue with status, owner and need-by filters; overdue rows in red	Open, Assign to me
6	scrFeasibility	APPG	Sourcing options side by side; reuse search of past requests	Add option, Select option, Generate report, Send for funding approval

#	Screen	Who sees it	What it shows	Main buttons
7	scrApprovals	Cost Center Owners	Requests waiting for my approval, with amount and report link	Open report (the decision itself is made in the Outlook or Teams card)
8	scrExecution	APPG	Milestones, progress %, expected completion date, issues	Start execution, Add milestone, Add issue, Save progress
9	scrClosure	APPG	Final cost, outcome, lessons learned, reuse recommendation, final documents	Close request
10	scrSearchHistory	Everyone (APPG sees all)	Search past requests by TVID, title, TV type or institution	Copy as Repeat Request

The Admin tile on scrHome does not open a screen. It opens the TVRMS Admin app in a new tab.

Model-driven app TVRMS Admin

Area	Pages (tables)	Who	What they do there
Master Data	Cost Centers, Research Institutions, APPG Stakeholders	APPG Admin	Add, edit and deactivate (never delete)
Settings	TVRMS Settings	APPG Admin	Change SLA days without a new release; every change is audited
Requests	TV Requests (all), Feasibility Options, Execution Milestones, TV Issues	APPG Admin, System Admin	Find any record and open its Audit history
Logs	Status History, Approval History, Notification Log	APPG Admin, System Admin	See who did what, and which emails went out

Screen design rules

- The same header on every screen: Lam logo, app title, user name and role, and a Home icon.
- Tablet layout with Scale to fit on; Lam colours and logo from the client.
- A button is visible only when the person's role allows the action, and enabled only when the request status allows it.
- Every long action shows a spinner, then a clear success or error message.
- Mandatory fields show a red star, and the Submit button explains what is missing.

8. Flows and notifications

Nine cloud flows do all the background work and send all 22 emails; the app only saves data. Because the flows react to changes in Dataverse, the rules still run when someone edits a row directly in TVRMS Admin.

The 9 flows

ID	Flow name	Trigger	What it does	Emails
F01	F01_NewRequestProcessing	Dataverse: a TV Request row is added	Folders, folder permissions, sharing with additional requesters, first audit row, first emails	N01, N02
F02	F02_StatusChangeEngine	Dataverse: a TV Request row is modified, column Request Status	Audit row, emails by status, execution guard, date stamps, RI Stakeholder access	N03, N04, N08, N09, N15, N16, N17, N19, N20

ID	Flow name	Trigger	What it does	Emails
F03	F03_FundingApproval	Dataverse: row modified, where Request Status = Pending Funding Approval and Approval Status = Not Started	Finds approvers in Cost Center, sends approval cards (Level 1, then Level 2 above the limit), approval history, result	N11, N14
F04	F04_MissingInfoDailyCheck	Schedule: daily 07:00	Reminders on day 7, 14 and 25; auto-cancel on day 30	N05, N06, N07
F05	F05_SLAReminders	Schedule: daily 07:30	Requests unassigned after 5 days; approval reminder on day 3, escalation from day 6	N12, N13, N21
F06	F06_ExecutionReminders	Schedule: daily 08:00	Milestones due in 2 days or overdue; overdue ones become Delayed	N18
F07	F07_WeeklyDigest	Schedule: Monday 09:00	One table of all open and overdue requests	N22
F08	F08_GenerateEvaluationReport	Called from the app (Power Apps V2), input TVID	Builds an HTML page, converts it to PDF, saves it in 02_Feasibility	N10

ID	Flow name	Trigger	What it does	Emails
F09	F09_UploadTVDocument	Called from the app (Power Apps V2), inputs TVID, Category, UploadedBy, File	Saves a file in the right TVID sub-folder	–

How flows and screens work together (example)

1. Priya clicks Mark Missing Information. The app changes only two things: Request Status and Missing Info Comments.
2. Dataverse saves the row. Flow F02 starts within seconds, because the Request Status column changed.
3. F02 writes the Status History row, sends N04 to Ravi with Priya's comment, logs the email and updates Previous Status.
4. Every morning F04 counts the days since Missing Info Since and sends a reminder on day 7, 14 or 25.

The 22 emails

Code	When	Sent by	To
N01	Request submitted	F01	Requester (cc additional requesters)
N02	New request waiting for review	F01	APPG team mailbox
N03	Owner assigned, feasibility started	F02	Requester (cc owner)
N04	More information needed	F02	Requester (cc additional requesters, owner)
N05, N06	Missing information reminder, day 7 and day 14	F04	Requester
N07	Final warning, day 25	F04	Requester (cc owner)

Code	When	Sent by	To
N08	Auto-cancelled after 30 days	F02	Requester (cc owner)
N09	Requester resubmitted	F02	APPG owner, or the APPG team if there is no owner yet
N10	Evaluation report ready	F08	APPG owner (cc requester)
N11	Approval card	F03	Level 1 or Level 2 approver
N12	Approval reminder, day 3	F05	Current approver
N13	Approval escalation, day 6 and later	F05	APPG team (cc current approver)
N14	Funding approved	F03	Requester (cc owner)
N15	Funding rejected	F02	Requester (cc owner)
N16	Sourcing on hold	F02	Requester (cc owner)
N17	Execution started	F02	Requester (cc owner, RI Stakeholder)
N18	Milestone due soon or overdue	F06	APPG owner (milestone owner)
N19	Request complete	F02	Requester (cc additional requesters, owner, approver)
N20	Request cancelled or rejected by a person	F02	Requester (cc owner)
N21	Requests not assigned for 5 or more days	F05	APPG team
N22	Weekly digest	F07	APPG team

The full email texts are in Chapter 9 of the build guide; use them as they are. Every email has an Open request button that opens the app directly on that request, for example ...&TVID=APTV_2026_014.

9. Build steps – Phase 0 (setup) and Phase 1 (data)

Phase 0 takes about 3 days and Phase 1 about 4 days, inside weeks 1 to 3. At the end, Dev has the solution, all 11 tables with sample data, the SharePoint library, and auditing switched on.

Phase 0 – Set up the base

Step 0.1 – Ask Lam IT for access (day 1). Send one email with this whole list, because some items take several days.

1. Three Power Platform environments with a Dataverse database: TVRMS-Dev, TVRMS-Test and TVRMS-Prod.
2. The System Customizer role for you in TVRMS-Dev (System Administrator is better in Dev).
3. Licences: Power Apps Premium for you and all app users, Power Automate Premium for the service account, Power BI Pro for you.
4. A service account svc-tvrms@lamresearch.com with a mailbox, added as a user in all 3 environments.
5. Six Entra ID security groups (Step 0.5), or permission to create them.
6. Three SharePoint sites (Dev, Test and Prod), or permission to create them.
7. A DLP check: Dataverse, SharePoint, Office 365 Outlook, Approvals, OneDrive for Business and Office 365 Users must be allowed together.

Step 0.2 – Environments. An environment is a separate space with its own data, apps and flows. Three environments keep testing away from real data.

Environment	Type	Purpose	Who uses it	SharePoint site
TVRMS-Dev	Sandbox	You build and change everything here	Developers	/sites/TVRMS-Dev
TVRMS-Test	Sandbox	UAT by business users	APPG team and a few requesters	/sites/TVRMS-Test
TVRMS-Prod	Production	Daily use	All Lam users	/sites/TVRMS

Rule: build only in Dev; Test and Prod only receive imports. Ask IT to link a security group to each environment, so only the right people can enter it.

Step 0.3 – Publisher and solution.

1. Open make.powerapps.com and choose TVRMS-Dev (top right).
2. Solutions → New solution. Display name: TV Request Management System. Name: TVRMS.
3. Publisher → New publisher. Display name: Lam APPG. Name: LamAPPG. Prefix: lam. Save and select it.
4. Version: 1.0.0.0 → Create.
5. From now on, create everything with + New inside this solution. Anything created outside it cannot move to Test or Prod.

Step 0.4 – Service account.

- Account svc-tvrms@lamresearch.com, display name TVRMS Notifications, with a Microsoft 365 licence and a mailbox.
- In each environment: Power Platform admin center → Environments → the environment → Settings → Users → add the account. Give it System Customizer now and the TVRMS Service role in Phase 4.
- Make it a Site Owner on the 3 SharePoint sites.
- All flows and connections belong to this account, so they keep running when a developer leaves.

Step 0.5 – Entra ID security groups. Entra admin center → Groups → New group → Group type: Security.

Group	Role	Sample members
TVRMS-Requesters	All Lam employees who may submit	Ravi Sharma, Meena Iyer
TVRMS-APPG-Owners	Review and run requests	Priya Nair, Arjun Mehta
TVRMS-APPG-Admins	Assign, hold, resume, master data	Anand Rao
TVRMS-CostCenter-Owners	Approvers (for access reviews and reports)	Suresh Kumar, Kavita Menon
TVRMS-RI-Stakeholders	Lam contacts for institutions	Rahul Verma

Group	Role	Sample members
TVRMS-System-Admins	Support team and the service account	svc-tvrms, support engineer

Step 0.6 – SharePoint site and libraries (documents only).

1. Create a private team site TVRMS-Dev. Site settings → Regional settings → the business time zone (default (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi; question G10).
2. Site permissions → sharing → Only site owners can share.
3. New → Document library → TV_Documents. Add the columns TVID, Document Category and Uploaded By Email (Section 6). Index TVID.
4. Library settings → Versioning settings → Create major versions, keep 100.
5. Create the library TV_Templates for the logo, the report template and the user guides.
6. Create the permission level Contribute No Delete: copy Contribute and untick Delete Items and Delete Versions. Create the SharePoint groups TVRMS APPG Owners, TVRMS APPG Admins and TVRMS Requesters, each with its Entra group inside. Write down the group IDs and the permission level ID (how: build guide 6.3 and 6.4).
7. TV_Documents → Stop inheriting permissions. Remove TVRMS Requesters (folder access comes from F01). Give the two APPG groups Contribute No Delete and the owners group Full Control.

Step 0.7 – Environment variables (technical values). Solution → New → More → Environment variable.

Name (schema name)	Type	Dev value (sample)	Used for
SiteURL (lam_SiteURL)	Text	https://lam.sharepoint.com/sites/TVRMS-Dev	Every SharePoint action
AppURL (lam_AppURL)	Text	Web link of the canvas app (fill after Phase 2)	Links in emails
AdminEmail (lam_AdminEmail)	Text	tvrms-support@lamresearch.com	Flow error alerts
APPGTeamEmail (lam_APPGTeamEmail)	Text	appg-team@lamresearch.com	N02, N13, N21, N22

Name (schema name)	Type	Dev value (sample)	Used for
SPGroupId_APPGOwners (lam_SPGroupId_APPGOwners)	Text	7	Folder permissions
SPGroupId_APPGAdmins (lam_SPGroupId_APPGAdmins)	Text	8	Folder permissions
RoleDefId_ContributeNoDelete (lam_RoleDefId_ContributeNoDelete)	Text	1073741924	Folder permissions

Business numbers such as SLA days go in the TVRMS Setting table instead. The APPG Admin can then change them with an audit trail and without a new release.

Step 0.8 – Connection references. Power Automate creates one per connector when you build a flow inside the solution: Dataverse, SharePoint, Office 365 Outlook, Approvals, OneDrive for Business and Office 365 Users. In Test and Prod you pick service-account connections during import.

Phase 1 – Build the data layer in Dataverse

Step 1.1 – Global choices. Solution → New → More → Choice. Create TVRMS Request Status, TVRMS Approval Status and TVRMS Role with the values in Section 6. Write down the number Dataverse shows for each option (for example 937520003), because flow filters use these numbers.

Step 1.2 – Create a table (worked example: Cost Center).

1. Solution → New → Table → Set advanced properties.
2. Display name Cost Center, plural Cost Centers. Primary column: Cost Center Code.
3. Advanced options → Record ownership: Organization. Tick Audit changes to its data. Save.
4. ◦ New column → Cost Center Name → Single line of text → Save.
5. ◦ New column → Level 1 Approver → Lookup → Related table: User → Save. Do the same for Level 2 Approver.
6. ◦ New column → Level 2 Threshold (USD) → Decimal with 2 places. Then Is Active → Yes/No, default Yes.
7. Edit the table data and type the 3 sample rows from Section 6, or use Import → Import data from Excel.

Repeat for Research Institution, APPG Stakeholder and TVRMS Setting; all four are Organization owned.

Step 1.3 – Create TV Request with the TVID autonumber.

1. New table TV Request (plural TV Requests). Primary column: Request Title. Record ownership: User or team. Tick Audit changes to its data.
2. ◦ New column → TVID → Data type Autonumber → Autonumber type Custom → Format APTV_{DATETIMEUTC:yyyy}_{SEQNUM:3} → Seed 1 → Save.
3. Add every other column from Section 6, Table 1. For Request Status choose Choice → Sync with global choice → TVRMS Request Status → default Submitted. Do the same for Previous Status (no default) and Approval Status (default Not Started).
4. Set Required = Business required on the requester's mandatory fields. The canvas form then shows a star and blocks Submit; flows are not blocked by this setting.

Example: the first request saved in 2026 gets APTV_2026_001; Ravi's request is the 14th, so it gets APTV_2026_014. Dataverse makes the number at save time, so two people can never get the same TVID.

Note: the number does not restart by itself in a new year. If Lam wants a restart at 001 (question G11), the System Admin resets the seed to 1 on 1 January, for example with the XrmToolBox Auto Number Manager.

Step 1.4 – Child tables and relationships.

1. Create Feasibility Option, Execution Milestone and TV Issue as User or team owned. Create Approval History, Status History and Notification Log as Organization owned.
2. In each child table add a Lookup column named TV Request, related table TV Request.
3. Open TV Request → Relationships → choose each relationship → Advanced options → Type of behavior: Parental for options, milestones and issues; Referential, restrict delete for the 3 audit tables.
4. Add the other lookups from Section 6: Parent Request, Cost Center, Selected Option, Selected Institution, Requester, Assigned APPG Owner and Current Approver.

Step 1.5 – Views. In TV Request → Views, create these 4 views for the APPG team and TVRMS Admin.

View	Filter	Sort
Open requests	Request Status is not Request Complete and not Request Cancelled	Created On, newest first
Overdue	Need By Date is before today, and the request is still open	Need By Date

View	Filter	Sort
Waiting for approval	Approval Status = Pending	Approval Requested On
Missing information	Request Status = Missing Information	Missing Info Since

Step 1.6 – Turn on auditing.

1. Power Platform admin center → TVRMS-Dev → Settings → Audit and logs → Audit settings. Turn on Start auditing. Set Retain these logs for to 7 years (or Forever).
2. Check that every TVRMS table has Audit changes to its data ticked (Steps 1.2 and 1.3).
3. Repeat in Test and Prod, because environment settings do not move with the solution.

Example: if someone changes the Funding Amount from 18,000 to 20,000 in TVRMS Admin, the row's Audit history shows the old value, the new value, who changed it and when. Audit logs use Dataverse log storage, so ask Lam IT to watch the capacity.

Step 1.7 – Sample master data. Load the sample rows from Section 6 into Cost Center, Research Institution, APPG Stakeholder and TVRMS Setting. Add a test cost center CC-TEST with your test users as approvers.

Checklist for Phase 0 and Phase 1

- ☐ 3 Dataverse environments ready; you have maker access in Dev
- ☐ Solution TVRMS with publisher prefix lam
- ☐ Service account licensed and added to all environments and sites
- ☐ 6 Entra groups with first members
- ☐ TV_Documents with versioning, the permission level and the SharePoint groups
- ☐ 7 environment variables created
- ☐ 3 global choices, 11 tables and all relationships with the right behaviour
- ☐ TVID tested: a test row gets APTV_2026_001
- ☐ Auditing on (environment and tables) with 7-year retention
- ☐ Sample master data and settings loaded

10. Build steps – Phase 2 (canvas app and admin app)

Phase 2 runs through Sprint 1 and Sprint 2 (weeks 3 to 6). Build the screens in the order of the lifecycle, and test each one with the sample request before you start the next. All formulas use global variables (Set); no collections are needed.

Step 2.1 – Create the canvas app and add data.

1. Solution TVRMS → New → App → Canvas app → Name TVRMS → Format Tablet → Create.
2. Settings → General (in some versions Updates) → turn on Automatically create environment variables when adding data sources. Do this before you add SharePoint, so the app finds the right site after import.
3. Data → Add data → Dataverse: TV Requests, Feasibility Options, Execution Milestones, TV Issues, Status Histories, Approval Histories, Cost Centers, Research Institutions, APPG Stakeholders, TVRMS Settings and Users.
4. Add SharePoint → the Dev site → TV_Documents. Add Office 365 Users for the people picker.
5. Settings → Data row limit = 2000. Keep Scale to fit on.

Step 2.2 – App.OnStart. Select App in the Tree view and paste this into OnStart.

```
// 1. Who is the user?
Set(varUserEmail, Lower(User().Email));
Set(varUserName, User().FullName);
Set(varMe, LookUp(Users, 'Primary Email' = User().Email));

// 2. Which role? (this only shows or hides tiles; Dataverse security is the
real lock)
Set(varStakeholder, LookUp('APPG Stakeholders', Email = varUserEmail && 'Is
Active' = true));
Set(varIsAdmin, varStakeholder.Role = 'TVRMS Role'. 'APPG Admin');
Set(varIsAPPG, varIsAdmin || varStakeholder.Role = 'TVRMS Role'. 'APPG
Owner');
Set(varIsApprover,
    !IsBlank(LookUp('Cost Centers',
        ('Level 1 Approver'.User = varMe.User || 'Level 2 Approver'.User =
varMe.User)
        && 'Is Active' = true)));

// 3. Settings from the TVRMS Setting table
Set(varMaxUploadMB, LookUp('TVRMS Settings', 'Setting Name' =
```

```

"MaxUploadMB").'Number Value');
Set(varAdminAppURL, LookUp('TVRMS Settings', 'Setting Name' =
"AdminAppURL").'Text Value');

// 4. Opened from an email link? (...&TVID=APTV_2026_014)
If(!IsBlank(Param("TVID")),
    Set(varSelectedRequest, LookUp('TV Requests', TVID = Param("TVID"))));

```

App → StartScreen: If(IsBlank(Param("TVID")), scrHome, scrRequestDetail)

Example: Priya signs in, so varIsAPPG = true and she sees the APPG Queue tile. Ravi signs in, all role variables are false, and he sees only New Request, My Requests and Search History.

Step 2.3 – Header. On scrHome insert a horizontal container conHeader: height 60, Fill RGBA(0, 63, 114, 1). Inside it put the Lam logo, the title TV Request Management System, a label with varUserName and the role, and a Home icon with OnSelect Navigate(scrHome). Copy conHeader to every new screen.

Step 2.4 – scrHome.

Control	Property	Formula
btnNewRequest	OnSelect	Set(varCopyFrom, Blank()); NewForm(frmNewRequest); Navigate(scrNewRequest)
btnAPPGQueue	Visible	varIsAPPG
btnMyApprovals	Visible	varIsApprover
btnAdmin	Visible / OnSelect	varIsAdmin / Launch(varAdminAppURL)
lblNeedMyAction	Text	"Need my action: " & CountRows(Filter('TV Requests', Requester.User = varMe.User && 'Request Status' = 'TVRMS Request Status':'Missing Information'))
lblWaitingApproval	Text	"Waiting for my approval: " & CountRows(Filter('TV Requests', 'Current Approver'.User = varMe.User && 'Approval Status' = 'TVRMS Approval Status'.Pending))

Control	Property	Formula
lblNewInQueue	Text (Visible: varIsAPPG)	"New requests: " & CountRows(Filter('TV Requests', 'Request Status' = 'TVRMS Request Status'.Submitted))

Step 2.5 – scrNewRequest.

1. Insert an Edit form frmNewRequest → Data source TV Requests → Default mode New → 2 columns. Add the fields: Request Title, Request Type, Parent Request, Frequency, Recurrence Details, TV Type, Description, Business Justification, Quantity, Need By Date, Cost Center, Estimated Budget (USD) and Additional Requesters.
2. Show fields only when needed. Rename the controls inside the Request Type and Frequency cards to ddRequestType and ddFrequency. Parent Request card → Visible and Required: ddRequestType.Selected.Value = 'Request Type (TV Requests)'.Repeat. Recurrence Details card → Visible and Required: ddFrequency.Selected.Value = 'Frequency (TV Requests)'.Recurring.
3. Cost Center control → Items: Filter('Cost Centers', 'Is Active' = true).
4. Additional Requesters: put a combo box cmbAddReq in the card, with multiple selection on. Items: Office365Users.SearchUserV2({searchTerm: Self.SearchText, top: 20}).value. Card Update: Concat(cmbAddReq.SelectedItems, Lower(Mail), ";").
5. Upload control: canvas apps have no stand-alone attachment control. Add a temporary Edit form on any SharePoint list, copy its Attachments control onto this screen, then delete the temporary form and data source. Name it attDocs and set Items = Blank(), MaxAttachments = 10 and MaxAttachmentSize = varMaxUploadMB.
6. Submit button btnSubmit → OnSelect:

```
// 1. Check the files
Set(varBadFile,
    CountRows(Filter(attDocs.Attachments,
        !(Lower>Last(Split(Name, ".")).Value) in
["pdf","docx","xlsx","pptx","png","jpg","zip","csv"]))) > 0);

If(
    !frmNewRequest.Valid,
        Notify("Please fill all mandatory fields (marked *).",
NotificationType.Error),
    CountRows(attDocs.Attachments) = 0,
        Notify("Please attach the Technical Documentation Package.",
NotificationType.Error),
```

```

varBadFile,
    Notify("File type not allowed. Use PDF, Word, Excel, PowerPoint, PNG,
JPG, ZIP or CSV.", NotificationType.Error),

// 2. All good: save the row (Dataverse gives the TVID now)
Set(varNewRequest,
    Patch('TV Requests', Defaults('TV Requests'), frmNewRequest.Updates,
        {
            Requester: varMe,
            'Request Status': 'TVRMS Request Status'.Submitted,
            'Approval Status': 'TVRMS Approval Status'.Not Started'
        }
    ));
// 3. Send each file to SharePoint through flow F09
ForAll(attDocs.Attachments As f,
    F09_UploadTVDocument.Run(varNewRequest.TVID, "01_Request",
varUserEmail,
    {contentBytes: f.Value, name: f.Name}));
    Notify("Request " & varNewRequest.TVID & " submitted. You will get an
email shortly.", NotificationType.Success);
    ResetForm(frmNewRequest); Reset(attDocs);
    Navigate(scrMyRequests)
)

```

Example: Ravi clicks Submit. The message says Request APTV_2026_014 submitted, and Wafer_Spec_v1.pdf lands in TV_Documents/APTV_2026_014/01_Request.

7. Copy as Repeat Request: each card's Default uses varCopyFrom when it is set.
 Request Title: If(IsBlank(varCopyFrom), ThisItem.'Request Title',
 varCopyFrom.'Request Title'). Request Type: If(IsBlank(varCopyFrom),
 ThisItem.'Request Type', 'Request Type (TV Requests)'.Repeat). Parent Request:
 If(IsBlank(varCopyFrom), ThisItem.'Parent Request', varCopyFrom).

Step 2.6 – scrMyRequests. Gallery galMyRequests → Items (ddStatus → Items:
 Choices('TVRMS Request Status')):

```
SortByColumns(
    Filter('TV Requests',
        (!varIsAPPG || Requester.User = varMe.User)
        && (IsBlank(ddStatus.Selected.Value) || 'Request Status' =
ddStatus.Selected.Value)
        && (IsBlank(txtSearch.Text) || StartsWith(TVID, txtSearch.Text)
        || StartsWith('Request Title', txtSearch.Text))
    ),
    "createdon", SortOrder.Descending)
```

Dataverse returns only the rows a person may see. So for Ravi this gallery shows his own requests plus those shared with him, with no extra filter. APPG users can see every row, so the first line keeps only their own.

Step 2.7 – scrRequestDetail. Screen OnVisible:

```
Set(varSelectedRequest, LookUp('TV Requests', 'TV Request' =
varSelectedRequest.'TV Request'));
Set(varIsMyRequest,
    varSelectedRequest.Requester.User = varMe.User
    || varUserEmail in Lower(varSelectedRequest.'Additional Requesters'));
Set(varTab, "Details");
```

Button	Visible when	What OnSelect does
btnAssignOwner	APPG, and status Submitted	Patch: Assigned APPG Owner = ddOwner.Selected.User, Request Status = Sourcing Feasibility In Progress
btnMissingInfo	APPG, and status Submitted or Sourcing Feasibility In Progress	Opens a pop-up; Patch: Request Status = Missing Information, Missing Info Comments, Missing Info Since = Now()
btnRejectReview	APPG, and status Submitted	Pop-up with reason; Patch: Request Status = Request Cancelled, Cancel Category = Rejected by APPG, Cancel Reason
btnEditResubmit	My request, and status Missing Information	EditForm(frmDetail); after saving, the status goes back (formula below)

Button	Visible when	What OnSelect does
btnCancel	My request while Submitted or Missing Information, or APPG Admin at any open status	Pop-up with reason; Patch: Request Status = Request Cancelled, Cancel Category, Cancel Reason
btnResume	APPG Admin, and status Sourcing On Hold	Patch: Request Status = Pending Funding Approval, Approval Status = Not Started; F03 starts again
btnUpload	My request or APPG, while the request is open	Pick a category, attach, then call F09
btnOpenFolder	Everyone who can see the request	Launch(varSelectedRequest.'Folder URL')

The main formulas:

```
// ddOwner.Items
Filter('APPG Stakeholders', 'Is Active' = true
    && (Role = 'TVRMS Role'. 'APPG Owner' || Role = 'TVRMS Role'. 'APPG
Admin'))

// btnConfirmMissing.OnSelect (inside the Missing Information pop-up)
If(IsBlank(txtMissingComments.Text),
    Notify("Please write what is missing.", NotificationType.Error),
    Set(varSelectedRequest, Patch('TV Requests', varSelectedRequest, {
        'Request Status': 'TVRMS Request Status'. 'Missing Information',
        'Missing Info Comments': txtMissingComments.Text,
        'Missing Info Since': Now() }));
    Set(varShowMissing, false);
    Notify("Requester informed.", NotificationType.Success))

// frmDetail.OnSuccess (after Edit and Resubmit)
Set(varSelectedRequest, Patch('TV Requests', frmDetail.LastSubmit, {
    'Request Status': If(IsBlank(frmDetail.LastSubmit.'Assigned APPG Owner'),
        'TVRMS Request Status'. Submitted,
        'TVRMS Request Status'. 'Sourcing Feasibility In Progress'),
    'Missing Info Since': Blank() }));
ViewForm(frmDetail);
Notify("Thank you. Your update was sent to the APPG team.",
NotificationType.Success)
```

```
// btnCancel.Visible
(varIsMyRequest && varSelectedRequest.'Request Status' in
  ['TVRMS Request Status'.Submitted, 'TVRMS Request Status'.'Missing
Information'])
|| (varIsAdmin && !(varSelectedRequest.'Request Status' in
  ['TVRMS Request Status'.'Request Complete', 'TVRMS Request
Status'.'Request Cancelled']))

// btnConfirmCancel.OnSelect (inside the cancel pop-up)
If(IsBlank(txtCancelReason.Text),
  Notify("Please enter a reason.", NotificationType.Error),
  Set(varSelectedRequest, Patch('TV Requests', varSelectedRequest, {
    'Request Status': 'TVRMS Request Status'.'Request Cancelled',
    'Cancel Category': If(varIsAdmin,
      'Cancel Category (TV Requests)'.'Cancelled by admin',
      'Cancel Category (TV Requests)'.'Cancelled by requester'),
    'Cancel Reason': txtCancelReason.Text }));
  Set(varShowCancel, false))
```

Tabs: Documents shows Filter(TV_Documents, TVID = varSelectedRequest.TVID). History (APPG only) shows Filter('Status Histories', 'TV Request'!'TV Request' = varSelectedRequest.'TV Request'), newest first. Approval shows the same filter on Approval Histories.

Step 2.8 – scrAPPGQueue. Gallery Items:

```
SortByColumns(
  Filter('TV Requests',
    'Request Status' <> 'TVRMS Request Status'.'Request Complete'
    && 'Request Status' <> 'TVRMS Request Status'.'Request Cancelled'
    && (IsBlank(ddQStatus.Selected.Value) || 'Request Status' =
ddQStatus.Selected.Value)),
  "lam_needbydate", SortOrder.Ascending)
```

TemplateFill: If(ThisItem.'Need By Date' < Today(), RGBA(255, 235, 235, 1), Color.White), so overdue rows show in red. The Assign to me button patches Assigned APPG Owner = varMe and Request Status = Sourcing Feasibility In Progress.

Step 2.9 – scrFeasibility.

1. galOptions → Items: Filter('Feasibility Options', 'TV Request'!'TV Request' = varSelectedRequest.'TV Request').

2. frmOption on Feasibility Options: TV Request card Default = varSelectedRequest;
Assessed By card Default = varMe.
3. Select button inside galOptions → OnSelect (only one option may be selected):

```
ForAll(Filter(galOptions.AllItems, 'Is Selected') As o,
  Patch('Feasibility Options', o, {'Is Selected': false}));
Patch('Feasibility Options', ThisItem, {'Is Selected': true});
Set(varSelectedRequest, Patch('TV Requests', varSelectedRequest, {
  'Selected Option': ThisItem,
  'Selected Institution': ThisItem.'Research Institution',
  'Funding Amount (USD)': ThisItem.'Estimated Cost (USD)' })))
```

4. Reuse search: galReuse → Items: Filter('TV Requests', 'Request Status' = 'TVRMS Request Status'.'Request Complete' && StartsWith('Request Title', txtReuse.Text)). It shows the outcome, cost and reuse recommendation of past requests.
5. btnReport → OnSelect: Set(varReport, F08_GenerateEvaluationReport.Run(varSelectedRequest.TVID)); Notify("Report saved in 02_Feasibility.", NotificationType.Success)
6. btnSendApproval → DisplayMode: enabled only when Selected Option and Report Generated On are filled. OnSelect: Patch Request Status = Pending Funding Approval and Approval Status = Not Started. Flow F03 starts by itself.

Example: Priya selects Option A, so the request now shows IMEC and USD 18,000. She generates the report and sends it, and Suresh gets the approval card within a minute.

Step 2.10 – scrApprovals. Items: Filter('TV Requests', 'Current Approver'.User = varMe.User && 'Approval Status' = 'TVRMS Approval Status'.Pending). Each row shows TVID, title, amount and cost center, plus an Open report button: Launch(ThisItem.'Folder URL' & "/02_Feasibility"). The decision itself is made in the Outlook or Teams card.

Step 2.11 – scrExecution.

```
// btnStartExecution.DisplayMode
If(varSelectedRequest.'Approval Status' = 'TVRMS Approval Status'.Approved
  && varSelectedRequest.'Request Status' = 'TVRMS Request Status'.'Pending
  Funding Approval',
  DisplayMode.Edit, DisplayMode.Disabled)
```

OnSelect patches Request Status = Execution In Progress and Expected Completion Date = dpExpected.SelectedDate. Below it, add a gallery and form for Execution Milestones and another for TV Issues (TV Request default = varSelectedRequest), and a slider

sldProgress (0 to 100) whose Save button patches Execution Progress (%).

Step 2.12 – scrClosure. btnClose → OnSelect:

```
If(
    IsBlank(txtActualCost.Text) || IsBlank(txtOutcome.Text) ||
    IsBlank(ddReuse.Selected),
        Notify("Actual cost, outcome and reuse recommendation are
mandatory.", NotificationType.Error),
        CountRows(Filter(TV_Documents, TVID = varSelectedRequest.TVID
&& 'Document Category'.Value = "05_Closure")) = 0,
        Notify("Please upload the final report into 05_Closure first.",
NotificationType.Error),
        Set(varSelectedRequest, Patch('TV Requests', varSelectedRequest, {
'Actual Cost (USD)': Value(txtActualCost.Text),
Outcome: txtOutcome.Text,
'Lessons Learned': txtLessons.Text,
'Reuse Recommendation': ddReuse.Selected.Value,
'Request Status': 'TVRMS Request Status'.Request Complete' }));
        Notify("Request closed.", NotificationType.Success)
    )
```

Step 2.13 – scrSearchHistory. Items: requests whose TVID or Request Title starts with the search text, plus a TV Type filter. Copy as Repeat Request → OnSelect: Set(varCopyFrom, ThisItem); NewForm(frmNewRequest); Navigate(scrNewRequest). Requesters find only their own requests (question G13); APPG find all.

Step 2.14 – Model-driven app TVRMS Admin.

1. Solution → New → App → Model-driven app → Name TVRMS Admin.
2. Navigation: add 4 groups. Master Data: Cost Centers, Research Institutions, APPG Stakeholders. Settings: TVRMS Settings. Requests: TV Requests, Feasibility Options, Execution Milestones, TV Issues. Logs: Status Histories, Approval Histories, Notification Logs.
3. For each table, put the key fields at the top of the main form and add useful columns to the Active view.
4. Save → Publish. Put the app link into the TVRMS Setting row AdminAppURL.
5. Share the app with the security roles TVRMS APPG Admin and System Administrator.

Example: Anand opens TVRMS Admin → Cost Centers → New → CC-7020, Thin Film R&D, Level 1 Approver Daniel Lee, threshold 50,000 → Save. The Audit history shows that Anand added it, and when.

Step 2.15 – Publish and share. Save and publish the canvas app. Share it with TVRMS-Requesters, TVRMS-APPG-Owners and TVRMS-APPG-Admins. Copy its web link into the AppURL environment variable (in Dev as the current value; in Test and Prod at import).

Checklist for Phase 2

- ☐ 10 screens built and tested with the sample request
- ☐ Buttons appear by role and status, exactly as in Sections 4 and 5
- ☐ No delegation warnings on formulas over TV Requests
- ☐ TVRMS Admin app with 4 areas, forms and views
- ☐ Both apps shared; AppURL and AdminAppURL filled

11. Build steps – Phase 3 (flows) and Phase 4 (security)

Build F01 and F02 first, because every other part depends on them. Then build F03, F08, F09 and the scheduled flows. Finish with the security roles and teams, and test with one user per role.

Rules for every flow

1. Create it inside the TVRMS solution: New → Automation → Cloud flow → Automated, Instant or Scheduled.
2. In Dev you may use your own connections. In Test and Prod, import while signed in as svc-tvrms, so the service account owns the flows and connections.
3. Rename every action with a clear name, for example Get cost center. Expressions use these names with underscores: outputs('Get_cost_center').
4. Put all steps in a Scope named Try. Add a Scope named Catch with Configure run after = has failed and has timed out. Inside Catch, email AdminEmail with the link to the failed run (build guide 8.2).
5. After every email, add a Notification Log row: Subject, TV Request, Notification Code, Sent To, Sent On = utcNow(), Delivery Status = Sent.
6. Read SLA days from the TVRMS Setting table with List rows. Never type SLA numbers inside a flow.
7. Dataverse filters use logical names in lowercase with the prefix, for example lam_requeststatus. To fill a lookup, type the table's set name and the row ID, for example lam_tvrequests(@{triggerOutputs()?['body/lam_tvrequestid']}).

F01 – New Request Processing

Trigger: Dataverse – When a row is added, modified or deleted. Change type: Added. Table name: TV Requests. Scope: Organization.

1. Get a row by ID → Users → Row ID: the Requester value from the trigger. This gives Ravi's email and full name. Do the same for Cost Center to get its code and name.
2. Apply to each over
["01_Request","02_Feasibility","03_Approval","04_Execution","05_Closure"] → SharePoint Create new folder → Site Address: SiteURL → Library: TV_Documents → Folder Path: concat(triggerOutputs()?['body/lam_tvid'], '/', item()).
3. Folder permissions with Send an HTTP request to SharePoint: break inheritance on the TVID folder, give the two APPG SharePoint groups Contribute No Delete, the owners group Full Control, and the requester Read. The exact calls are in build guide 8.3 Part E; use only the folder part, because the request row now lives in Dataverse.
4. Apply to each email in split(Additional Requesters, ';') → List rows → Users → Filter rows: internalemailaddress eq '<email>' (rename it Find user) → Perform an unbound action → GrantAccess. The action shows the expected input format when you select it; the values look like this:

Target:

```
lam_tvrequests(@{triggerOutputs()?['body/lam_tvrequestid']})
```

PrincipalAccess:

```
{
  "Principal": {
    "@odata.type": "Microsoft.Dynamics.CRM.systemuser",
    "systemuserid": "@{first(outputs('Find_user')?['body/value'])?['systemuserid']}"
  },
  "AccessMask": "ReadAccess, WriteAccess, AppendAccess, AppendToAccess"
}
```

Then give the same person Read on the TVID folder (the same HTTP call as in step 3).

5. Update a row (TV Requests, this row): Folder URL = concat(SiteURL, '/TV_Documents/', TVID) and Previous Status = Submitted.
6. Add a new row → Status Histories: Name = TVID, TV Request = lam_tvrequests(<row id>), New Status = Submitted, Changed By = the requester, Changed On = utcNow(), Comments = Request submitted.

7. Send N01 (to the requester, cc the additional requesters) and N02 (to APPGTeamEmail), each followed by a Notification Log row.

Example run: Ravi submits at 10:30. By about 10:31 the folder APTV_2026_014 has its 5 sub-folders, Meena can open the request, and N01 and N02 are in the inboxes.

F02 – Status Change Engine

Trigger: Dataverse – row Modified, table TV Requests, scope Organization. Select columns: lam_requeststatus, so the flow runs only when the status changes.

1. Compose New = Request Status and Old = Previous Status, both from the trigger.
2. Condition: New equals Old → Terminate (Succeeded). This skips F01's own update and any save that did not really change the status.
3. Execution guard: if New = Execution In Progress and Approval Status is not Approved, update the row back to Old, email the APPG team "Execution blocked for <TVID>: funding is not approved", and Terminate. The guard's own update starts F02 once more, and step 2 ends that run.
4. Add a Status History row: Old Status, New Status, Changed By = Modified By, Changed On = utcNow(), and Comments (the Missing Info Comments or Cancel Reason when relevant).
5. Switch on New:

New status	What F02 does	Email
Sourcing Feasibility In Progress	From Submitted: stamp Feasibility Started On (if empty)	N03 from Submitted; N09 from Missing Information
Missing Information	–	N04 with the comments
Submitted (coming from Missing Information)	–	N09 to the APPG team
Pending Funding Approval	Nothing; F03 does the work	–
Sourcing On Hold	–	N16
Execution In Progress	Stamp Execution Start Date; GrantAccess Read for the RI Stakeholder of the Selected Institution; Read on the folder	N17

New status	What F02 does	Email
Request Complete	Stamp Closed On	N19
Request Cancelled	Stamp Closed On; pick the email by Cancel Category	N08 (Auto-cancelled), N15 (Funding rejected), N20 (all others)

6. Last step, always: Update a row → Previous Status = New. This update does not touch Request Status, so it does not start F02 again.

F03 – Funding Approval

Trigger: row Modified, table TV Requests. Select columns: lam_requeststatus, lam_approvalstatus. Filter rows: lam_requeststatus eq <number of Pending Funding Approval> and lam_approvalstatus eq <number of Not Started>.

1. Get the Cost Center row (Level 1 Approver, Level 2 Approver, Level 2 Threshold), then both approvers' user rows (email and name).
2. Update the request: Approval Status = Pending, Current Approver = the Level 1 approver, Approval Requested On = utcNow().
3. GrantAccess (ReadAccess) on the request for the approver, plus Read on the TVID folder.
4. Start and wait for an approval. Type: Custom responses – Wait for one response. Response options: Approve, Reject, Hold. Title: Funding approval: APTV_2026_014 – USD 18,000. Assigned to: the approver's email. Details: the N11 text with links to the report and the app. Item link: the AppURL with &TVID=<TVID>.
5. Action settings → Timeout: P25D (25 days; a flow run cannot wait longer than 30 days). Treat a timeout as Hold.
6. Add an Approval History row: level, approver, decision, comments, date and amount.
7. If the outcome is Approve and the Funding Amount is above the Level 2 Threshold, repeat steps 2 to 6 for Level 2.
8. Set the result:

Outcome	F03 sets	Email
Approve (all levels)	Approval Status = Approved, Approved On = utcNow()	N14 (sent by F03)

Outcome	F03 sets	Email
Reject	Approval Status = Rejected; Request Status = Request Cancelled; Cancel Category = Funding rejected; Cancel Reason = the comment	N15 (sent by F02)
Hold or timeout	Approval Status = On Hold; Request Status = Sourcing On Hold	N16 (sent by F02)

Example: the approval for APTV_2026_014 reaches Suresh in Outlook and in the Teams Approvals app. He approves with OK within budget; 18,000 is below 50,000, so there is no Level 2, and Ravi gets N14.

F04 to F07 – scheduled flows

Flow	Schedule	Rows it reads	What it does
F04	Daily 07:00	TV Requests with Request Status = Missing Information	Days = whole days since Missing Info Since. Day 7: N05. Day 14: N06. 5 days before the cancel day: N07. On or after the cancel day (30): Request Status = Request Cancelled and Cancel Category = Auto-cancelled; F02 then sends N08
F05	Daily 07:30	Submitted requests older than SLA_ReviewDays; requests with Approval Status = Pending	One combined N21 to the APPG team; day 3 of an approval: N12 to the approver; day 6 and later: N13 to the APPG team (cc approver)
F06	Daily 08:00	Execution Milestones that are not Done	Due in 2 days: N18 due soon. Overdue: Milestone Status = Delayed and N18 overdue (on the first day, then weekly)
F07	Monday 09:00	All open TV Requests	HTML table with TVID, title, status, owner, need-by date and an Overdue flag: N22

Days between two dates: $\text{div}(\text{sub}(\text{ticks}(\text{utcNow}()), \text{ticks}(<\text{the date}>)), 864000000000)$, because one day is 864000000000 ticks. Set every Recurrence trigger to the business time zone (question G10).

Example: Ravi is asked for information on 08-Oct and does nothing. He gets reminders on 15-Oct, 22-Oct and 02-Nov, and the request is cancelled automatically on 07-Nov.

F08 – Evaluation report (PDF)

1. Trigger: Power Apps (V2) with one Text input, TVID.
2. List rows → TV Requests → Filter rows: `lam_tvid eq '<TVID>'`, Row count 1. List rows → Feasibility Options → Filter rows: `_lam_tvrequest_value eq <request id>`.
3. Compose the HTML page: request details, a comparison table of the options, then the recommendation (template in build guide 8.10).
4. OneDrive for Business → Create file `/TVRMS_Temp/<TVID>.html` → Convert file to PDF.
5. SharePoint → Create file in `TV_Documents/<TVID>/02_Feasibility`, named `<TVID>_Evaluation_Report.pdf`. The name never changes, so SharePoint keeps every version.
6. Update the request: Report Generated On = `utcNow()`. Delete the temporary OneDrive file.
7. Send N10, log it, and Respond to a Power App or flow with status = OK.

F09 – Upload TV document

1. Trigger: Power Apps (V2) with inputs in this order: Text TVID, Text Category, Text UploadedBy, File.
2. SharePoint → Create file → Folder Path `/TV_Documents/<TVID>/<Category>` → File Name and File Content from the File input. Create file makes missing folders in the path, so the order of F01 and F09 does not matter; test this once in Dev.
3. Update file properties: TVID, Document Category and Uploaded By Email.
4. Find the request by TVID and update it: Is Doc Uploaded = Yes.
5. Respond to a Power App or flow with status = OK.

The flow runs with the service account's connection, so requesters can add files even though they only have Read on their folder.

Phase 4 – Security, step by step

1. **Security roles.** Power Platform admin center → the environment → Settings → Users + permissions → Security roles. Select Basic User → Copy → name it TVRMS Requester, and set the privileges from Section 4. Repeat for TVRMS APPG Owner, TVRMS APPG Admin and TVRMS Service.
2. **Add the roles to the solution.** Solution → Add existing → Security role → the 4 roles. They then move to Test and Prod with the solution.
3. **Teams.** Settings → Users + permissions → Teams → Create team. Team type: Microsoft Entra ID Security Group; choose the group; Membership type: Members. Create TVRMS Requesters, TVRMS APPG Owners, TVRMS APPG Admins and TVRMS System Admins, and give each team its security role. Teams do not move with the solution, so repeat this in every environment.
4. **Service account.** Give svc-tvrms the TVRMS Service role; it needs the Share privilege for GrantAccess.
5. **Share the apps.** Canvas app → Share → the 3 Entra groups. TVRMS Admin → share with the roles TVRMS APPG Admin and System Administrator.
6. **SharePoint.** Library permissions were set in Step 0.6; flows F01, F02 and F03 add folder access per request.
7. **Optional hardening (suggestion S7).** Column security on Approval Status and Funding Amount (USD), so only APPG and the service account can change them.
8. **Test with one user per role** (Section 12, TC05 to TC07).

Checklist for Phase 3 and Phase 4

- ☐ 9 flows built inside the solution, each with Try and Catch
- ☐ F01 tested: TVID folder, permissions, sharing, emails, audit row
- ☐ F02 tested for every status in the Switch, including the execution guard
- ☐ F03 tested at USD 18,000 (one level) and USD 60,000 (two levels), plus Reject and Hold
- ☐ Scheduled flows tested by running them by hand with back-dated rows
- ☐ 4 security roles in the solution; 4 teams linked to Entra groups
- ☐ One test user per role passes TC05 to TC07

12. Testing and UAT

Testing runs in three rounds. You test every screen and flow in Dev while you build; you run the whole process in Test after the first import; then Lam business users run UAT in Test and sign off. All cases use the sample data from Section 2.

Test accounts (one per role)

Test user	Role	Set-up
Test Requester A	Requester	TVRMS-Requesters only
Test Requester B	Additional requester, or another requester	TVRMS-Requesters only
Test APPG Owner	APPG Owner	TVRMS-APPG-Owners + APPG Stakeholder row (APPG Owner)
Test APPG Admin	APPG Admin	TVRMS-APPG-Admins + APPG Stakeholder row (APPG Admin)
Test Approver L1	Cost Center Owner	Level 1 Approver of test cost center CC-TEST
Test Approver L2	Finance Head	Level 2 Approver of CC-TEST, threshold 50,000
Test RI Stakeholder	RI Stakeholder	RI Stakeholder of a test institution

Test cases

ID	Area	Steps	Expected result
TC01	Intake	Submit without a document	Error message; nothing saved
TC02	Intake	Submit with an .exe file	Error: File type not allowed

ID	Area	Steps	Expected result
TC03	Intake	Submit a full valid request	TVID such as APTV_2026_001 in the success message; folder with 5 sub-folders; file in 01_Request; N01 and N02; Status History row
TC04	TVID	Two users submit at the same second	Two different TVIDs, no duplicate
TC05	Security	Requester B opens the app, TVRMS Admin and SharePoint	Cannot see Requester A's request or folder
TC06	Security	Requester A adds B as additional requester	B sees the request and gets cc on N01
TC07	Security	Any business user tries to delete a row or a file	Delete is not possible
TC08	Review	APPG assigns an owner	Status Sourcing Feasibility In Progress; N03; Feasibility Started On filled; history row
TC09	Missing info	APPG marks Missing Information	N04 with the comments; Missing Info Since filled
TC10	Missing info	Requester edits and resubmits	Status goes back; N09; Missing Info Since empty
TC11	Auto-cancel	Set Missing Info Since to 31 days ago in TVRMS Admin; run F04 by hand	Request Cancelled; Cancel Category Auto-cancelled; N08
TC12	Reminders	Set Missing Info Since to exactly 7 days ago; run F04	Only N05
TC13	Reject at review	APPG rejects a Submitted request with a reason	Request Cancelled; Rejected by APPG; N20
TC14	Feasibility	Add 2 options and select one	Only one Is Selected; Selected Institution and Funding Amount filled

ID	Area	Steps	Expected result
TC15	Report	Generate the report twice	PDF in 02_Feasibility with 2 versions; N10; Report Generated On filled
TC16	Approval	Send USD 18,000 for approval; L1 approves	One card; history row; Approval Status Approved; N14; approver can open the request
TC17	Approval	Send USD 60,000	Two cards in order (L1 then L2); two history rows
TC18	Approval	Approver rejects with a comment	Request Cancelled; Funding rejected; N15
TC19	Approval	Approver holds; APPG Admin resumes	Sourcing On Hold and N16; after resume, a new card
TC20	Guard	Set Execution In Progress in TVRMS Admin before approval	Status put back; alert email to the APPG team
TC21	Execution	Start execution after approval	N17 to requester and RI Stakeholder; RI Stakeholder can open the request
TC22	Milestones	Milestone planned 2 days ahead; run F06	N18 due soon; an overdue milestone becomes Delayed
TC23	Closure	Close without actual cost, then with all data	First an error; then Request Complete, Closed On and N19
TC24	Repeat	Copy an old request as Repeat	Form pre-filled; Request Type Repeat; Parent Request set
TC25	Deep link	Open the link in an email	The app opens on that request
TC26	Audit	Change Funding Amount in TVRMS Admin	Audit history shows old value, new value, who and when
TC27	Settings	Change SLA_ReviewDays from 5 to 3	F05 uses 3 next morning; the change shows in the audit history
TC28	Reports	Refresh Power BI	Counts match the Dataverse views

ID	Area	Steps	Expected result
TC29	Flow errors	Break a connection in Test and trigger a flow	The Catch email reaches the support mailbox

UAT plan

Item	Plan
Environment	TVRMS-Test, never Prod
When and how long	3 to 5 working days: the end of week 6 and the start of week 7
Participants ("limited users" in the deck)	APPG lead, 2 APPG owners, 1 APPG admin, 2 requesters, 1 cost center owner
Entry criteria	TC01 to TC29 passed in Test; master data loaded; test users ready; a 1-hour demo done
Daily routine	Users test in the morning; a 15-minute call at 5 PM to review new defects
Defect severity	Critical: process stops, no workaround. High: wrong result, workaround exists. Medium: small issue. Low: text or layout
Exit criteria	No Critical or High defects open; all scenarios passed; written sign-off received

UAT scenario (end to end)

1. The requester submits 10 coupons for adhesion test, CC-TEST, USD 8,000, with a spec PDF. Check: the TVID email arrives.
2. The APPG Admin assigns an APPG Owner. Check: the requester gets N03.
3. The APPG Owner marks Missing Information: Add drawing. The requester uploads the drawing and resubmits. Check: N04 and N09.
4. The APPG Owner adds 2 options, selects one, generates the report and sends it for approval. Check: the PDF exists and the approver gets the card.
5. The approver approves. The APPG Owner starts execution and adds 2 milestones. Check: N14 and N17.

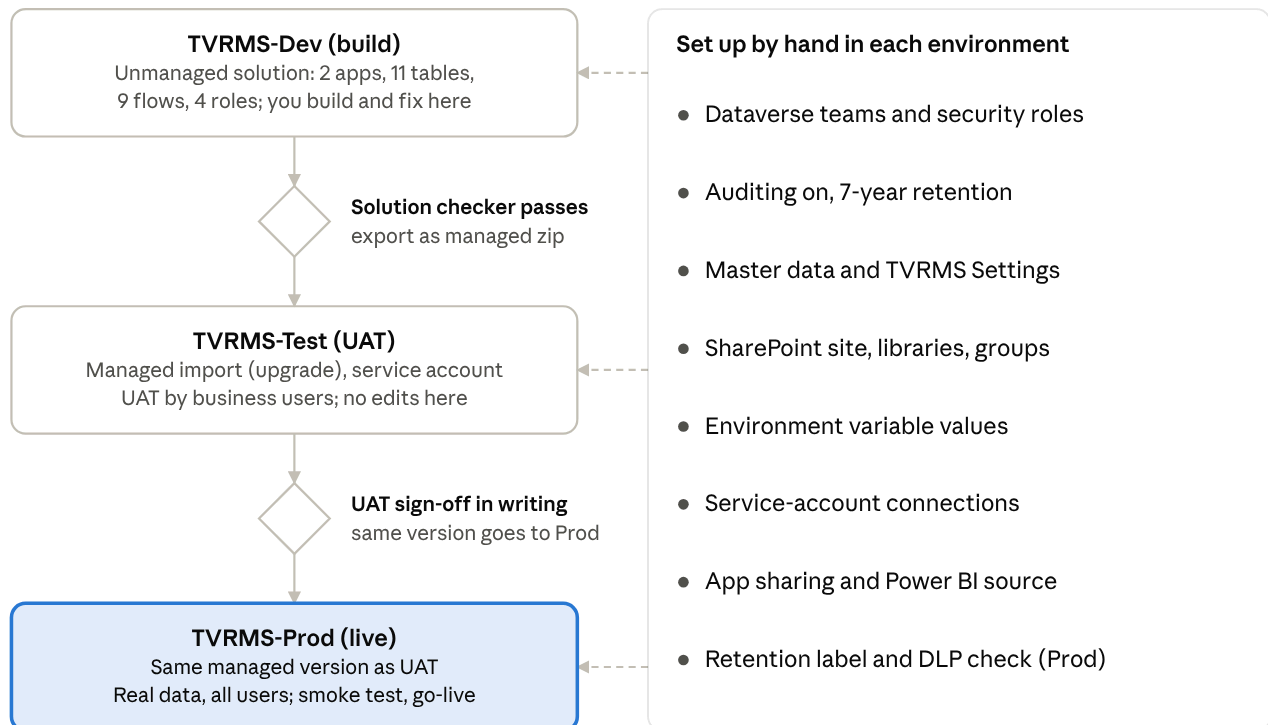
6. The APPG Owner closes the request with cost 7,600 and an outcome. Check: N19, status Request Complete, and Power BI shows it after refresh.

Keep a defect log with these columns: Defect ID, Date, Raised by, Screen or Flow, Steps, Expected, Actual, Severity, Status (Open, Fixed, Retest, Closed), Fixed in version. For the sign-off email, use the text in build guide 11.7.

13. Deployment to production and hypercare

The path is always Dev → Test → Prod. You build in Dev, export the solution as managed, and import the same version into Test for UAT and then into Prod. Some parts move inside the solution; others you set up by hand in each environment.

One managed version moves Dev → Test → Prod; nobody edits Test or Prod



deployment path · Dev, Test and Prod with 2 gates

Only the managed solution travels between environments, and only after it passes its gate. The right panel lists what you set up by hand in every environment; fix problems in Dev and import a new version as an upgrade, never by editing Test or Prod.

What moves with the solution, and what you set up by hand

Moves inside the solution	Set up by hand in each environment
Canvas app TVRMS and model-driven app TVRMS Admin	The environment, its Dataverse database and its security group
11 tables, columns, relationships, views, forms, global choices and the TVID autonumber	Dataverse teams linked to Entra groups, with their security roles
9 flows	Auditing on, with 7-year retention
4 security roles	Master data and TVRMS Settings rows (real values in Prod)
Environment variable definitions and connection references	Environment variable values and connections (asked during import)
–	SharePoint site, TV_Documents and TV_Templates, permission level and SharePoint groups
–	Power BI data source and refresh, Purview retention label, DLP check, app sharing

Steps

1. **Export from Dev.** Solution → Publish all customizations. Raise the version (for example 1.0.0.3). Run Solution checker and fix every High issue. Export as Managed. Also export an Unmanaged copy, dated, as your source backup.
2. **Prepare Test.** Create the SharePoint site and libraries, the permission level and the groups, and note the new IDs. Create the 4 teams with their roles. Turn on auditing. Load the test master data and settings.
3. **Import into Test.** Sign in as svc-tvrms → Solutions → Import solution → the managed zip. Choose service-account connections. Enter the Test values for the environment variables (site URL, group IDs, role ID, emails). Import, then check that all 9 flows are On. Share the apps, fill AppURL and AdminAppURL, and run the Section 2 story yourself before UAT starts.
4. **UAT and fixes.** Fix only in Dev. Export a new version and import it into Test as an Upgrade. Never edit a managed solution in Test or Prod.

5. **Production.** Do Steps 2 and 3 again with Prod values, using the exact version that passed UAT. Ask Lam IT to apply the 7-year Purview retention label to TV_Documents and to check the DLP policy. Load the real master data. Share the apps with the Entra groups.
6. **Smoke test.** Submit one dummy request, take it to Request Complete, and check every email. This request uses a real TVID (for example APTV_2026_001) and stays in the system, so tell Lam in advance and give it a clear title such as Go-live smoke test.
7. **Reports and go-live mail.** Point Power BI to the Prod environment, publish it and schedule the refresh. Send the go-live email with the app link and the user guide.

Go-live checklist

- ☐ UAT sign-off received in writing
- ☐ Managed solution in Prod is the same version that passed UAT; all 9 flows On
- ☐ Flows and connections owned by the service account
- ☐ Environment variable values point to Prod (site URL, group IDs, role ID, emails, AppURL)
- ☐ Teams linked to Entra groups with the right security roles; tested with one real requester
- ☐ Auditing on with 7-year retention; Purview retention label on TV_Documents
- ☐ Master data and TVRMS Settings checked by the APPG Admin
- ☐ DLP policy checked
- ☐ Power BI published from Prod, refresh scheduled, access limited to APPG and management
- ☐ Support mailbox receives flow error alerts
- ☐ Go-live email sent with the app link and the user guide
- ☐ Hypercare plan agreed (weeks 8 and 9)

Hypercare (weeks 8 and 9) and handover

- Every morning, check the run history of the 9 flows; fix and resubmit failed runs.
- Check the Notification Log for emails with Delivery Status = Failed.
- Hold a daily 15-minute call with the APPG lead.
- Collect small changes in a backlog and release them together as version 1.1, through Dev → Test → Prod.

- At the end of week 9, hand over to the Lam application support team: this guide, the solution files and version history, the list of environment variables and settings, the flow catalogue (Section 8), the admin and user guides (build guide Chapter 14), the open defect list, and a recorded knowledge-transfer session.

Support level	Who	Handles
L1	APPG Admin	How-to questions, master data, reassignments, hold, resume and cancel
L2	Lam application support (HCLTech during hypercare)	Failed flows, permissions, app defects, small changes
L3	Microsoft, through Lam IT	Platform outages and service issues

14. Gaps to ask the client

The two documents are good enough to start the build, but not to finish it. Below are 5 decisions where the documents disagree and 29 business questions. Each one has a suggested default, so the build never waits: if Lam does not answer within 3 working days, we build with the default and confirm it by email.

A. Decisions (the two documents disagree)

#	Decision	Question for Lam	Why it matters	Our suggestion
D1	Data store	Do all app users have, or will they get, Power Apps Premium, so we can use Dataverse as the deck shows?	Dataverse gives row-level security, audit and scale, but every app user needs a premium licence	Dataverse for data, SharePoint for documents. Fallback without premium: the SharePoint-list design in build guide Chapters 5 and 6, with the same screens and flows

#	Decision	Question for Lam	Why it matters	Our suggestion
D2	Admin app	Is a model-driven admin app (deck) fine, instead of an admin screen inside the main app?	Model-driven apps need Dataverse and look different from the canvas app	Model-driven app TVRMS Admin for master data, settings and logs
D3	Timeline	Is go-live at the end of week 7 still the target, with UAT inside weeks 6 and 7?	The build guide estimated 10 weeks to go-live for one developer	Keep 9 weeks; if UAT finds blockers, go live with the MVP first (Section 16)
D4	Migration and training	Please confirm that old requests are not migrated and that HCLTech does not train business users (deck: out of scope)	The build guide planned both; together they take 1 to 2 weeks	Out of scope; we hand over the user guides; migration only as a change request
D5	TVID workspace	Does workspace per TVID mean a folder (our plan) or a separate SharePoint site per request?	A site per request is heavy to create, secure and retain	One folder per TVID with 5 standard sub-folders

B. Business questions

#	Topic	Question for Lam	Suggested default
G1	Form fields	Which fields are on the request form, and which are mandatory?	The requester fields in Section 6, Table 1; all mandatory except Additional Requesters, Parent Request (Repeat only) and Recurrence Details (Recurring only)
G2	TV types	Which TV types exist?	Wafer, Coupon, Material, Module, Other

#	Topic	Question for Lam	Suggested default
G3	Technical documents	Which documents are mandatory at submission? Allowed file types and size?	At least 1 file (the Technical Documentation Package); PDF, Word, Excel, PowerPoint, PNG, JPG, ZIP, CSV; 10 MB per file in the app
G4	Internal alignment	Who takes part in the 1st and 2nd internal alignment? Do they need their own status or sign-off?	The APPG Owner runs both rounds inside Sourcing Feasibility In Progress; no extra status
G5	Reject at review	Can APPG reject a request at review? Should Rejected be a separate status?	Yes; Request Cancelled with the category Rejected by APPG and a reason, which keeps 8 statuses
G6	Approval matrix	How many approval levels? Any amount limits?	Level 1 = Cost Center Owner, always; Level 2 = Finance Head above USD 50,000 (limit stored per cost center)
G7	Reject and Hold	What happens after Reject or Hold?	Reject: Request Cancelled. Hold: Sourcing On Hold until the APPG Admin resumes or cancels
G8	Approver away	Who approves when the cost center owner is on leave?	The approver reassigns in the Approvals app, or the APPG Admin changes the Cost Center row and resumes
G9	SLA days	How many days for review, reminders, escalation and auto-cancel?	Review 5; approval reminder day 3, escalation day 6; missing-info reminders day 7, 14 and 25; auto-cancel day 30; milestone reminder 2 days before
G10	Calendar and time zone	Working days or calendar days? Which time zone for scheduled emails?	Calendar days; IST (UTC+05:30), or Pacific Time if the APPG team is in the US

#	Topic	Question for Lam	Suggested default
G11	TVID	Is APTV_YYYY_NNN right? Must it restart at 001 each year?	Yes; Dataverse autonumber, seed reset to 1 every 1 January
G12	Evaluation report	Is there a template or logo? Should the requester get it?	Standard PDF from the system with the Lam logo; the requester is copied on N10
G13	Requester visibility	Can requesters see the feasibility comparison, the audit, or all past requests?	They see only their own requests: status, approval result and documents
G14	RI Stakeholders	Who are they? Do external institution contacts need access?	One Lam employee per institution, with read access from execution start; no external access
G15	Recurring requests	Is each run a new request?	Yes; each run gets its own TVID through Copy as Repeat Request
G16	Cancel rights	Who can cancel, and when?	The requester while Submitted or Missing Information; the APPG Admin at any open status; a reason is always required
G17	Execution exceptions	What counts as an exception or issue? Who resolves it? Does it need approval?	TV Issue rows with type, severity, owner, target date and resolution; overdue milestones become Delayed
G18	Cost and utilization	What cost and utilization tracking is needed: per request, per institution, per cost center?	Estimated vs. funded vs. actual cost per request; spend and request count per institution and per cost center in Power BI
G19	Currency	Are all amounts in USD?	USD only, stored as decimal numbers

#	Topic	Question for Lam	Suggested default
G20	Notifications	Is the list of 22 emails right? Which mailbox sends them? Teams messages too?	The 22 emails from the build guide, sent from svc-tvrms or a tvrms-noreply shared mailbox; approvals also show in Teams
G21	Report audience	Who may see the Power BI dashboard? Do they all have Power BI Pro?	APPG team and management only; Pro licences from Lam (deck assumption)
G22	KPIs	Which KPIs and targets: cycle time, aging buckets, approval rate?	Status counts, cycle time per phase, aging 0-7 / 8-30 / over 30 days, cost variance, approval and completion rates
G23	Volume and users	How many requests and app users per year?	Up to 500 requests a year; the user count decides the licence count
G24	Environments	When will Dev, Test and Prod with Dataverse, the licences and the service account be ready?	Ready in week 1 (a deck dependency)
G25	Retention	Who applies the 7-year Purview label? What happens after 7 years? Who watches audit log storage?	Lam IT applies the label and watches storage; nothing is deleted automatically in phase 1
G26	Branding and language	Languages, colours and logo?	English only; Lam logo and colours from the client
G27	Master data	Who provides cost centers, approvers, institutions and APPG people, in which format, and by when?	Lam fills our Excel template by the end of week 2
G28	Product owner	What does APPG stand for, and who is the APPG lead who signs off?	Needed for the Solution Report and the UAT sign-off

#	Topic	Question for Lam	Suggested default
G29	Devices	Do users need the app on phones or inside Teams?	Desktop browser (Edge); the app can also be added as a Teams tab

How to use these tables

1. On day 1, send the email below to the business owner and the APPG lead, with both tables attached.
2. Add a column Lam answer and fill it as answers come in.
3. No answer in 3 working days: send a short follow-up, "If we do not hear back by <date>, we will build with the defaults." Keep that email as written confirmation.
4. When an answer differs from the default, change only that setting, column or flow step, and note it in the change log.

Ready-to-send email

Subject: TVRMS - 5 decisions and 29 questions before the build (defaults included)

Dear <Business Owner>, dear <APPG Lead>,

Thank you for the TV Request Management System scope. To build it right, we need your decision on 5 points and answers to 29 questions (tables attached). For each one we have written the default we will use if we do not hear back by <date + 3 working days>.

The 5 decisions:

1. Licences: do all app users have Power Apps Premium? (needed for Dataverse)
2. Admin app: a model-driven app for master data, settings and logs - OK?
3. Timeline: UAT in weeks 6-7 and go-live at the end of week 7 - still the target?
4. Please confirm that data migration and business-user training are out of scope.
5. "Workspace per TVID" = one SharePoint folder per request with 5 sub-folders - OK?

We also need the master data in the attached Excel template, the name of the APPG lead who will sign off, and access to Dev, Test and Prod by <date>.

Best regards,
Syed
Technical Lead, HCLTech

15. Assumptions and suggestions

Assumptions are what we take as true to plan and price the work. If one proves false, HCLTech and Lam assess the effect on time and cost together. Suggestions are small improvements we propose; Lam can accept or drop each one.

Assumptions

#	Assumption	Source	If it proves false
A1	Lam provides the master data (users, departments, cost centers, locations, vendors) in the required format	Deck	We build with sample data and swap in the real data later; go-live waits for it
A2	Lam stakeholders join workshops, reviews, approvals and decisions, and give feedback on time	Deck	The plan slips; raise it with Lam's project sponsor
A3	The timeline is re-checked in Plan for Success (weeks 1 and 2), and gaps are closed together	Deck	Re-plan at the end of week 2
A4	Power BI licences are available to dashboard users	Deck	Users without a licence use the Excel export from the views
A5	Business users do UAT	Deck	Go-live waits for a written sign-off
A6	Anything not listed in scope is out of scope	Deck	A change request, with its impact on cost and time
A7	Lam provides environments, access and all licences from day 1	Deck	The build cannot start; days lost are added to the plan

#	Assumption	Source	If it proves false
A8	No integration with external systems, such as purchasing or ERP	Deck	A change request
A9	Lam owns data migration, business-user training and change management	Deck	A change request
A10	Up to 500 requests a year; English only	Build guide	The design still works; more users means more licences
A11	Lam IT applies the retention label, the DLP policy and the environment security groups; HCLTech has no tenant admin rights	Build guide	Those steps wait for Lam IT
A12	Email is the main channel; approvals also appear in Teams	Build guide	Teams chat messages would be a small change request

Suggestions to propose

S1 to S5 are already part of this guide's design, so Lam only needs to agree to them. S6 to S10 are optional extras.

#	Suggestion	Benefit for Lam	Effort
S1	TVID as a Dataverse autonumber column	Never duplicates, and the number is there the moment the request is saved	Smaller than a counter flow
S2	TVRMS Setting table for SLA days and limits	The APPG Admin changes rules without a new release, and every change is audited (an NFR in the deck)	Small
S3	TV Issue table for exceptions and issues	Covers the deck's exception management and exception analysis	Small
S4	Cancel Category column	Clear exception reports: rejected vs. auto-cancelled vs. withdrawn	Very small

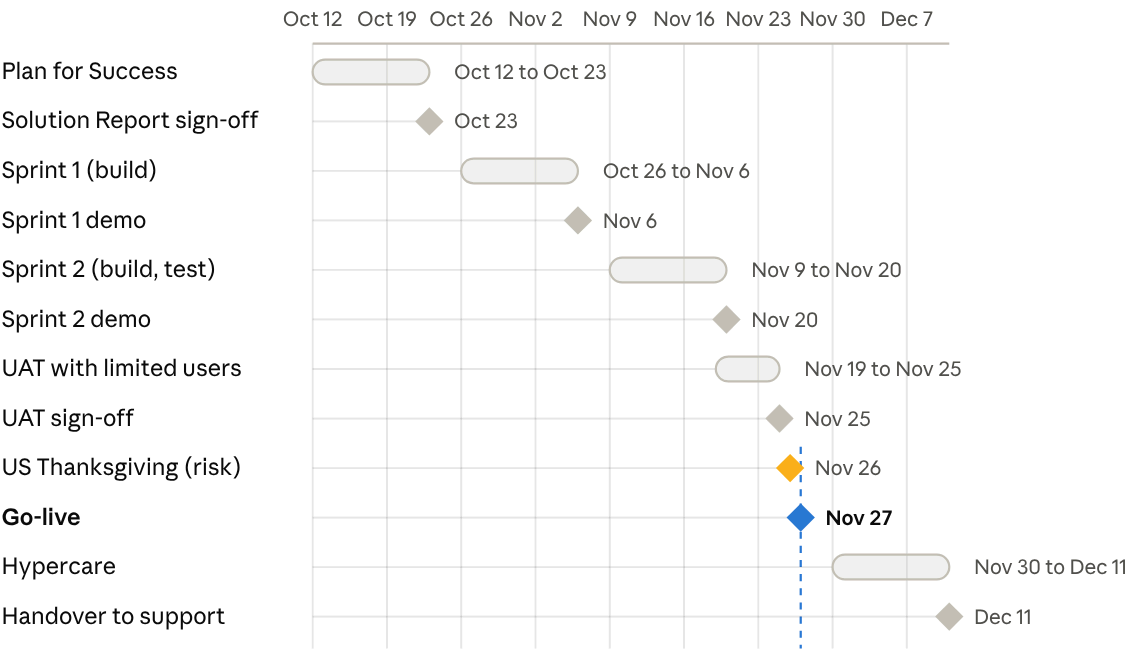
#	Suggestion	Benefit for Lam	Effort
S5	Reuse Recommendation at closure	Covers the deck's reuse recommendations and feeds the reuse search	Very small
S6	Add the app as a Teams tab	Users find it where they already work; approvals already arrive in Teams	Very small
S7	Column security on Approval Status and Funding Amount	Only APPG and the flows can change the approval and money fields	Small
S8	Power Platform Pipelines for Dev → Test → Prod	Repeatable, logged deployments (needs Managed Environments)	Small
S9	MVP fallback	If time runs short, go live with the core process first (Section 16)	Planning only
S10	Monthly access review	A list of who holds which role, ready for audits	Small

16. Project plan: 9 weeks

The plan follows the deck: 2 weeks to plan, two 2-week sprints to build and test, 1 week to deploy and go live, and 2 weeks of hypercare. Sample dates assume a start on Monday 12-Oct-2026; shift them to the real start date.

Go-live lands at the end of week 7, the day after US Thanksgiving

9-week plan with sample dates, starting Monday 12-Oct-2026



project timeline · 9 weeks, sample dates

Bars are phases and diamonds are milestones, both placed from the plan rows. If Lam’s US users are away for Thanksgiving week, move go-live to Tuesday 1-Dec and shift hypercare by 2 working days.

Week by week

Week (sample dates)	Phase	Work	Demo or deliverable
W1 (12–16 Oct)	Plan for Success	Kick-off; send the gap email (Section 14) and the access request (Step 0.1); walk through today’s manual process, pain points and reporting needs	Access requested; decisions D1 to D5 under discussion
W2 (19–23 Oct)	Plan for Success	Business rules confirmed; solution blueprint; Phase 0 set-up; data model drafted in Dev	Solution Report submitted for approval

Week (sample dates)	Phase	Work	Demo or deliverable
W3 (26–30 Oct)	Sprint 1	Tables and sample data; F01 and F02; Home, New Request, My Requests and Request Detail	A submitted request gets its TVID, folder, emails and audit row
W4 (2–6 Nov)	Sprint 1	APPG Queue, Missing Information, Reject at review; F04; security roles and teams; TVRMS Admin	Sprint 1 demo: intake and review cycle
W5 (9–13 Nov)	Sprint 2	Feasibility screen, F08 report, F03 approval, My Approvals	Report and approval card in Outlook and Teams
W6 (16–20 Nov)	Sprint 2	Execution, Closure and Search History; F05, F06, F07, F09; Power BI; system test; first import to Test; UAT starts on Thursday	Sprint 2 demo: the full process in Test
W7 (23–27 Nov)	Deploy	UAT ends; fixes; sign-off on Wednesday; Prod set-up; go-live on Friday	Signed-off apps; go-live
W8–W9 (30 Nov–11 Dec)	Hypercare	Daily checks, fixes, the version 1.1 backlog, knowledge transfer	Handover to Lam application support

Watch out: Thursday 26-Nov-2026 is US Thanksgiving, and Lam is a US company. Confirm that US users and approvers are available that week, or move go-live to Tuesday 1-Dec and shift hypercare by 2 working days.

Deliverables (from the deck)

Phase	Deliverable	What it contains
Plan for Success	Solution Report	Confirmed requirements, answers to D1 to D5 and G1 to G29, the design in Sections 3 to 8, and the plan

Phase	Deliverable	What it contains
Design, Build and Test	Signed-off Power Platform applications	The solution (2 apps, 11 tables, 9 flows, 4 roles), the Power BI report, test results and the UAT sign-off
Hypercare	Handover to support	This guide, admin and user guides, the flow catalogue, version history, open defects and a recorded knowledge-transfer session

Your first 5 days

1. **Day 1:** send the gap email (Section 14) and the access request (Step 0.1). Book the kick-off and the process walk-through.
2. **Day 2:** as soon as Dev is ready, create the publisher, the solution, the global choices and the environment variables.
3. **Day 3:** create the 4 master tables with sample rows, and the SharePoint library TV_Documents.
4. **Day 4:** create TV Request with the TVID autonumber, the 6 child tables and the relationship behaviours.
5. **Day 5:** build F01, test it by adding a row in the table editor (Tables → TV Request → Edit), and show the result (TVID, folder, emails) to your manager.

If time runs short: the MVP

Agree this fallback with Lam before week 6, not in the middle of UAT.

Release	What goes live	When
MVP	Intake, TVID, folders, APPG review, missing information, feasibility, report, funding approval, execution and closure (F01, F02, F03, F04, F08, F09)	End of week 7
Version 1.1	Reminders and escalations (F05, F06), the weekly digest (F07) and the Power BI dashboard	During hypercare, weeks 8 and 9

17. How to present this to the client

Use a 30-minute walkthrough, and show pictures rather than tables. Spend 5 minutes on the problem, 10 on the story of one request, 5 on the architecture, 7 on the decisions and 3 on the plan.

The 30-minute walkthrough

Minutes	Show	Say (simple words)
0–5	Section 1, Today vs. with TVRMS	"Today TV requests live in emails, slides and spreadsheets. TVRMS gives every request one ID, one folder, one status and a full history."
5–15	Section 2 story, then the lifecycle picture in Section 5	"Let me follow one request: Ravi's 25 wafers, from submission on 7 October to closure on 12 December."
15–20	The architecture picture in Section 3	"Everything stays inside Lam's Microsoft 365: Entra ID for sign-in, Dataverse for data, SharePoint for files, Outlook and Teams for approvals, Power BI for reports."
20–27	Decisions D1 to D5 in Section 14	"We need 5 decisions this week. For each one we already have a recommendation, so the build can start now."
27–30	The timeline picture in Section 16	"Nine weeks: 2 to plan, 4 to build in 2 sprints, 1 to go live, 2 of hypercare. Here is what we need from you, and by when."

The one-minute version

"TVRMS is one app for every Test Vehicle request. The requester fills one form and attaches the technical document. The system gives the request a TVID, creates its folder and sends it to the APPG team. APPG compares sourcing options, and the cost center

owner approves the money in Outlook or Teams. Every step sends an email, every change is logged, and nothing is deleted for 7 years. Managers see status, cycle time and cost in Power BI."

Questions Lam may ask

Question	Answer
Can a requester see other people's requests?	No. Dataverse security shows only their own requests and the ones shared with them.
What if the approver is on leave?	The approver can reassign in the Approvals app, or the APPG Admin changes the approver and resumes.
Can we change the reminder days later?	Yes. The APPG Admin changes them in TVRMS Admin, and the change is logged.
Are records ever deleted?	No. Nobody has delete rights; records and files are kept for at least 7 years.
Can we add a field later?	Yes. We add it in Dev and release a new version through Test to Prod.
Does it work on a phone?	The app runs in Edge and other approved browsers; approvals also work in Outlook and Teams on mobile.
What licences do our users need?	Power Apps Premium for app users and Power BI Pro for report viewers; approvers who only use Outlook or Teams need nothing extra (decision D1).

Tips for the meeting

- Talk in phases and business words. Keep flows, tables and formulas for the technical session.
- For each decision, ask for one named owner and a date for the answer.
- Never promise anything from the out-of-scope list; say "we can add it as a change request".

- Close by confirming the next 2 dates: the Solution Report review (end of week 2) and the Sprint 1 demo (end of week 4).